

DEPARTMENT OF AERONAUTICS
2019 - 2020 GROUP DESIGN PROJECT

Wholly Reusable Launch System: Launch System Team

**Regenerative Cooling, Pressurisation and Fluid Flow
Control for a LOX/LCH₄ Expander Cycle Rocket Engine**



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Álvaro Fieira Wachtendorff, June 2020

Abstract

As part of the *Wholly Reusable Launch System* Group Design Project, the propulsion sub-team was in charge of designing *Mantis*, the engine which will serve *Astraea*, a rocket whose mission requirements will be to take a 160 kg payload to 800 km orbit and be fully reusable.

During the conceptual design stage, several trade-off analyses were carried out and an expander cycle liquid bi-propellant rocket engine was selected as the most feasible reusable option. In addition to this, the regenerative cooling system required to extract the waste heat of the system and bring the materials to working limits was designed, consisting of 34 distinct coolant ducts. The total power extraction and the pressure losses across the cooling jacket were quantified: 279.1126 kW and 6.8584 bar respectively.

Furthermore, devices for fluid flow control, such as control ball and globe valves, were selected and a regression data analysis was performed to size the pipelines throughout the rocket. Finally, the configuration of the pressurisation system was finalised by selecting helium as the pressurant, and its mass was calculated: 14.8273 kg.

Contents

Acknowledgements	i
Abstract	ii
Contents	iii
List of Tables	iv
List of Figures	v
Nomenclature	vi
1 Conceptual Design	1
1.1 Team Strategy	1
1.2 Number of Stages	1
1.3 Choice of Propulsive System	1
1.4 Choice of Engine Cycle	2
1.5 Choice of Propellant	2
2 Regenerative Cooling	3
2.1 Introduction	3
2.2 Pressure Drops Across Regenerative Cooling System	3
2.3 Axial Variations and Power Extraction	4
2.4 Manufacturability	6
2.5 Design of Coolant Channels	6
3 Fluid Flow Control and Pressurisation	8
3.1 Rocket Lines	8
3.2 Material Choices for Lines	9
3.3 Choice of Valves	9
3.4 Pressurisation and Helium Mass Requirements	11
3.5 Discussion	12
A Raw Data	13
A.1 CEA Lower Stage	13
A.2 CEA Upper Stage	14
A.3 Data from NASA's Rocket Lines	14
B MATLAB Code	16
Bibliography	35

List of Tables

1.1	Number of stages required for the rocket's budget ΔV .	1
2.1	Prices obtained from CES EduPack [1].	6
2.2	Final values for the design of the regenerative cooling system.	7
3.1	Selected diameters for <i>Astraea</i> 's pipelines.	8
3.2	Material properties for 321 CRES and Inconel 718. *Cost relative to 6 ft x 3 in of 6061-T6, data acquired from [2],[3] and [4].	9
3.3	Helium mass required, all values in kilograms.	12
A.1	Raw data obtained from NASA's <i>Rocket Lines</i> [2], used for pipeline sizing. Pressure and temperature conditions omitted.	15

List of Figures

1.1	Number of stages required.	1
1.2	Simple schematic view of an expander cycle, created by the author.	2
2.1	Diagram of a regenerative cooling system, created by the author.	3
2.2	Methane phase diagram, values obtained from the Engineering Toolbox [5].	3
2.3	Sectional pressure drops across the nozzle.	4
2.4	Cumulative pressure drop across the nozzle.	4
2.5	Diagram showing the heat transfer from the inside of the nozzle to the coolant channels, created by the author.	4
2.6	Axial variation of adiabatic wall temperature.	5
2.7	Axial variation of heat transfer coefficient.	5
2.8	Axial variation of the total heat transferred.	5
2.9	Ashby charts obtained from CES EduPack [1].	6
2.10	Channel design for a constant mass flow of 2 kg/s.	7
2.11	Channel design for a constant thickness of 3 mm.	7
3.1	Bellow dimensions from regression data, scaled down taking into account thrust capabilities and engine dimensions.	8
3.2	Flexible hose dimensions from regression data, scaled down taking into account thrust capabilities and engine dimensions.	8
3.3	Ball valve design, created by the author.	9
3.4	Flow coefficient variations with pressure drop for a ball valve.	9
3.5	Schematic layout of <i>Mantis</i> showing the pressure drops across the system, created by the author. . . .	10
3.6	Globe valve design, created by the author.	10
3.7	Flow coefficient variations with pressure drop for a globe valve.	10
3.8	Flow control loop, created by the author.	11
3.9	Ullage inside propellant tanks, created by the author.	11
3.10	Schematic layout of pressurisation system, created by the author.	12

Nomenclature

β	Payload Factor m_{pl}/m_{oN}	h_c	Coolant-Side Heat Transfer Coefficient [Wm ⁻² K ⁻¹]
ΔV	Delta V [ms ⁻¹]	h_g	Gas-Side Boundary Layer Heat Transfer Coefficient [Wm ⁻² K ⁻¹]
γ	Ratio of Specific Heats C_p/C_v	h_v	Latent Heat of Vaporisation of Propellant [Jkg ⁻¹]
μ	Dynamic Viscosity [kgm ⁻¹ s ⁻¹]	I_{sp}	Specific Impulse [s]
σ	Correction Factor for h_g	k	Thermal Conductivity [Wm ⁻¹ K ⁻¹]
M	Mach Number	K_v	Flow Coefficient [m ³ hr ⁻¹ bar ^{-0.5}]
Nu	Nusselt Number $h_c d/k$	l	Section Length [m]
Pr	Prandtl Number $C_p \mu/k$	m_f	Final Mass [kg]
Re	Reynold's Number $\rho u D/\mu$	m_o	Initial Mass [kg]
ε	Structural Factor m_{fN}/m_{oN}	M_v	Mass of Vaporised Propellant [kg]
A	Section Area [m ²]	N	Number of Stages
A_t	Throat Area [m ²]	p	Pressure [Pa]
c^*	Characteristic Velocity [ms ⁻¹]	Q	Flow Rate [m ³ hr ⁻¹]
C_p	Specific Heat Capacity [Jkg ⁻¹ K ⁻¹]	q	Heat Flux [Wm ⁻²]
C_{pl}	Specific Heat Capacity of Liquid Propellant [Jkg ⁻¹ K ⁻¹]	R	Specific Gas Constant [Jkg ⁻¹ K ⁻¹]
C_{pv}	Specific Heat Capacity of Vaporised Propellant [Jkg ⁻¹ K ⁻¹]	r	Local Recovery Factor, $Pr^{2/3}$ for Turbulent Flows
d	Coolant Channel Diameter [m]	SG	Specific Gravity
D_t	Section Diameter [m]	t_w	Wall Thickness [m]
f_D	Darcy-Weisbach's Friction Factor	T_{aw}	Adiabatic Wall Temperature [K]
G	Payload to Liff-Off Mass and Coolant Mass per Unit Area [kgm ⁻²]	T_{co}	Coolant Flow Temperature [K]
g_e	Gravitational Acceleration [ms ⁻²]	T_e	Propellant Tank Temperature [K]
H	Experimental Heat Transfer Coefficient [Jm ⁻² K ⁻¹ s ⁻¹]	T_u	Ullage Tank Temperature [K]
		T_v	Vaporised Propellant Temperature [K]
		T_{wc}	Coolant-Side Wall Temperature [K]
		T_{wg}	Gas-Side Wall Temperature [K]
		V_{He}	Volume of Pressurant [m ³]
		V_T	Volume of Tank [K]

Chapter 1

Conceptual Design

In a world where the demand for space flight is on an exponential growth curve, the need for reusable systems to rapidly cheapen launch costs has lead to the creation of *Astraea*. Named after the ancient Greek for star, *Astraea* will be a rocket capable of launching a payload of up to 160 kg to Low Earth Orbit, at an altitude of 800 km and be fully reusable. *Mantis*, named after the predatory insect capable of jumping 45 times its height, is the propulsive solution to *Astraea*. It will be a liquid bi-propellant expander cycle rocket engine, which will render this project possible.

This report will take the reader through several sections of the design procedure of *Mantis*, providing the justifications for any decisions made during the conceptual design and detailing the methods employed during the preliminary and detailed designs.

1.1 Team Strategy

The first two weeks of this project were dedicated to the conceptual design and were carried out collaboratively between the four members of the Propulsion Sub-Team. The first steps of the conceptual design lied in the calculation of the number of stages required, and the selection of the type of rocket, engine cycle and propellant needed to meet this project's requirements. This section will aim to provide the reasoning behind the decisions that were made during this first stage.

During the preliminary and detailed designs, the following roles were allocated.

- Álvaro Fieira Wachtendorff - Cooling System, Fluid Flow Control and Pressurisation.
- Kieran Jones-Tett - Propellant Analysis, Nozzle Geometry, Reaction Control System.
- Sabapathy Tirupathi Ramana - Turbomachinery and On-Board Power Generation.
- Tham Ju Rong, Ryan - Cost Analysis, Combustion Chamber.

1.2 Number of Stages

The initial step during the design process was to identify the number of stages required for the rocket. This problem was approached by finding the total ΔV *Astraea* will need, given by expression 1.1:

$$\Delta V = g_e I_{sp} \ln(m_o/m_f) \quad (1.1)$$

This equation can be employed for every stage and the total ΔV calculated from their summation. The mass fraction m_o/m_f for every stage can be expressed in terms of the structural (ε) and payload (β) factors, where the former is just the ratio between final and initial stage masses and the latter the ratio of payload mass to stage initial mass [6].

$$\Delta V = -N g_e I_{sp} \ln(\varepsilon (1 - \beta) + \beta) \quad (1.2)$$

The payload factor can be re-expressed as $G^{-1/N}$, where G is equal to the payload mass divided by the total lift off mass of the rocket. The optimal number of stages, 2, was found by plotting the relationship of G versus ΔV and using the parameters in table 1.1, provided by the Systems Sub-Team.

Table 1.1: Number of stages required for the rocket's budget ΔV .

G	118.275
Budget ΔV (m/s)	10,006
Number of stages	2

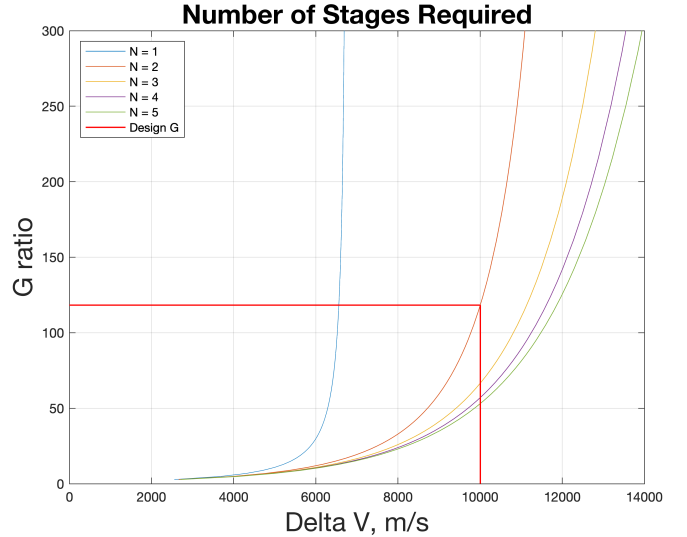


Figure 1.1: Number of stages required.

1.3 Choice of Propulsive System

Since reusability is the main driver of this project, the propulsive system of the *Astraea* had to be designed

taking into account the fact that full recovery of the first and second stages is required. Two main possible scenarios were identified. The first one was to land propulsively, with one or two re-entry burns, on an ASDS (Autonomous Spaceport Drone Ship) out at sea, precisely positioned at the location of stage touchdown, in a similar fashion to SpaceX's current plan of action [7].

Another simpler approach would rely on the structure's atmospheric deceleration, perhaps aided by a propulsive re-entry, to aerially recover the structure. This has been successfully achieved by Rocket Lab USA™, whose modus operandi involves a hook hanging from a helicopter that captures the first stage of the rocket after being decelerated by a parafoil during descent [8].

Either way, solid rocket motors were discarded due to their inability to reignite, were it to be necessary, and their low reusability. The core of every solid rocket motor, where the grain is stored, must be re-built after every flight [9]. Liquid motors, on the other hand, are highly reusable and reignitable, and can also provide throttling meaning that controllability of the rocket is enhanced during flight [6]. Therefore, a liquid bi-propellant motor was chosen for *Mantis*.

1.4 Choice of Engine Cycle

A liquid bi-propellant engine, as its name suggests, uses two main propellants: an oxidizer and a fuel, which when mixed generate enough thrust for lift-off. The engine cycle is the layout of the fluid flow within the rocket motor: the turbine drives the turbopumps which in turn drive the flow into the combustion chamber at high pressures. Throughout the years, several configurations have been developed. A common engine cycle is the gas-generator which makes use of a small pre-burner chamber where a small portion of the propellants are burned to generate the required power to drive the pumps. For this project nonetheless, the expander cycle was selected. Not only do they have a higher than average specific impulse and a simpler layout but are also the best option in terms of reusability [10]. Their compatibility with low temperature propellants leads to almost no damage in the system.[11].

Expander cycles make use of the waste heat from the combustion chamber and the nozzle to drive the turbine, which then drives the pumps which feed flow into the system, through the cooling jacket into the combustion chamber (see figure 1.2). The pressure that can be obtained in the combustion chamber is thus dependent on the heat extraction of the system, which in some situations might not be large enough. This explains why expander cycle engines exist only in the upper stages of rockets, where low chamber pressures are

acceptable [12]. Nevertheless, the low-scale nature of this project permits the use of more modest combustion chamber pressures. *Astraea* will therefore make use of 10 expander cycle *Mantis* engines, 9 for the lower stage and 1 for the upper stage, adapted to vacuum conditions by having a larger area ratio (A_e/A_t).

It is worth mentioning that one current lower stage expander cycle, the LE-9, is currently under development at JAXA [13].

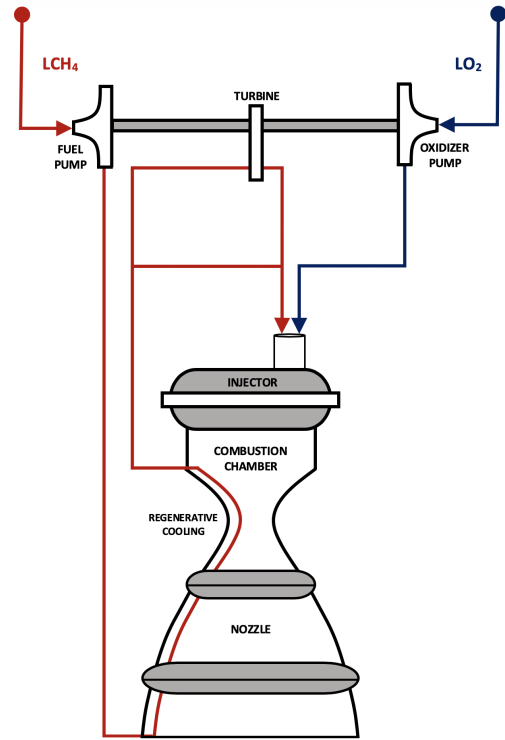


Figure 1.2: Simple schematic view of an expander cycle, created by the author.

1.5 Choice of Propellant

In order to extract enough heat for an expander cycle to be feasible, cryogenic propellants must be employed. This shortened down the list of possible fuel choices. Once again, for matters of reusability, liquid methane was chosen. Not only does it provide higher coking limits, but also experiences less soot deposition allowing for a longer life expectancy for the engine. In addition to this, methane's boiling point, 111K, is similar to that of liquid oxygen, allowing for a simpler tank architecture. Another advantage is the higher specific impulse in comparison with other very commonly used fuels like hydrogen and kerosene (RP-1) [14]. Kieran Jones-Tett's report includes a detailed comparison of the performance of methane and other propellants.

Chapter 2

Regenerative Cooling

2.1 Introduction

The backbone of any expander cycle is the cooling system, since not only does it allow for the walls of the combustion chamber and the nozzle to come within the material limits, but also extracts the waste heat from ignition and converts it to mechanical energy at the turbine. The turbine then drives the pumps which feed propellant flow into the cooling system.

This method of energy extraction is called regenerative cooling, and in order to calculate the amount of heat energy can be extracted, the principals of heat energy transfer must be understood.

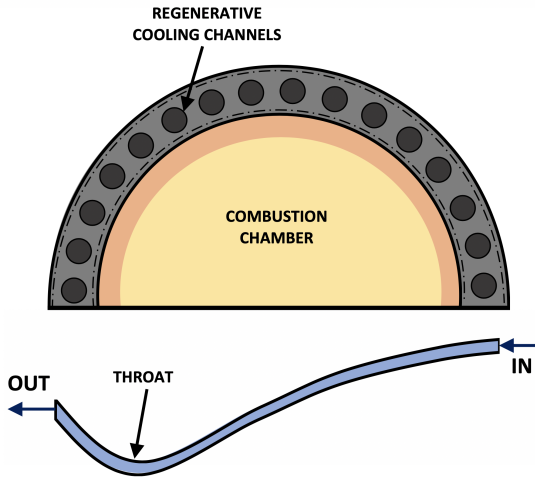


Figure 2.1: Diagram of a regenerative cooling system, created by the author.

2.2 Pressure Drops Across Regenerative Cooling System

Propellants flowing through the cooling jackets are often above critical pressure, which is the required pressure to liquefy a gas below its critical temperature. Gases above their critical pressure do not experience a sudden change of phase. Instead, they experience a gradual change from a liquid-like substance to a gas-like substance. The lack of sudden phase change means that energy extracted is used to increase the coolant's temperature instead of breaking molecular bonds. This eliminates the need to consider latent heats, easing the modelling of the heat transfer and increasing the efficiency of the system [15].

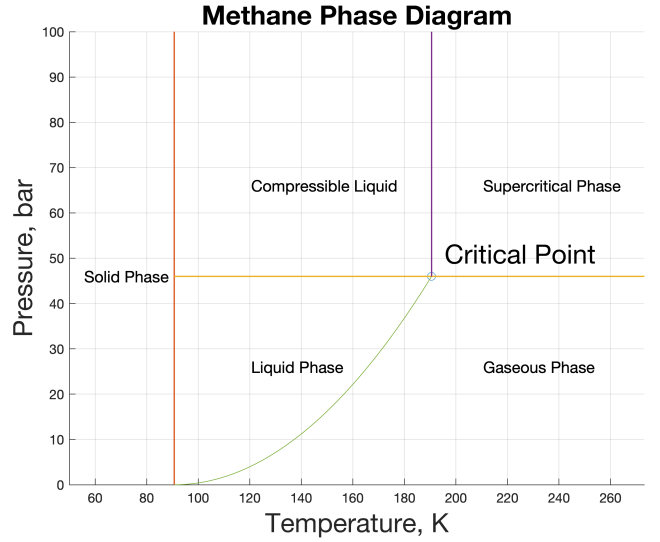


Figure 2.2: Methane phase diagram, values obtained from the Engineering Toolbox [5].

Methane has a critical pressure of 45.99 bar [5]. The inlet pressure of the coolants must be considerably greater than that of the combustion chamber to account for the pressure losses within the cooling system. It is due to this that methane will be expected to be utilised at pressure above critical, in a liquid or supercritical phase.

To calculate the pressure drop, the nozzle was discretised into 26 sections. The pressure drop across each section is given by Colebrook's equation [16]:

$$\Delta p = \frac{1}{2} f_D \frac{l}{D} \rho u^2 \quad (2.1)$$

where l is the section length, D its diameter, ρ the coolant's density, u its velocity and finally f_D , Darcy-Weisbach's friction factor which can be determined iteratively using expression 2.2 [17]:

$$\frac{1}{\sqrt{f_D}} = -2 \log_{10} \left(\frac{h_s}{3.7D} + \frac{2.51}{Re f_D} \right) \quad (2.2)$$

where h_s is the roughness height, and Re the Reynolds number at the considered section.

Figure 2.3 shows the pressure drop expected at every station of the nozzle. As is can be seen, the most significant pressure increase occurs just prior to the throat where the rate of change of curvature is greatest leading to a significant change in the friction factor. Note how the horizontal axis is not to scale: the subsonic region was augmented so as to facilitate the visualization of the results at the throat area.

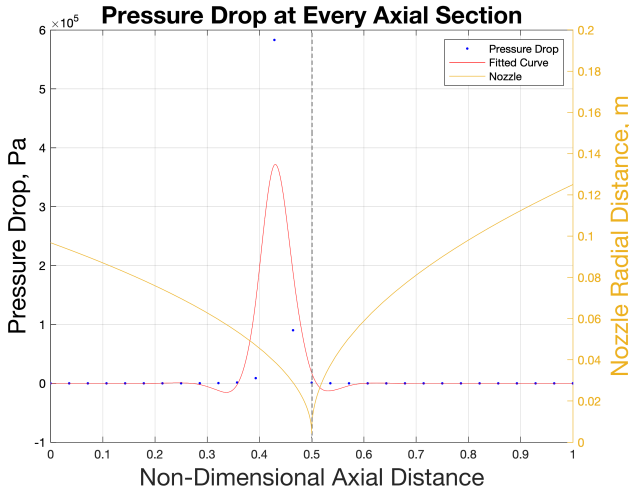


Figure 2.3: Sectional pressure drops across the nozzle.

Figure 2.4 shows the cumulative pressure drop across the nozzle from the right, the supersonic region, to the left, the subsonic region. The total pressure drop across the regenerative cooling system was determined to be 6.8584 bar.

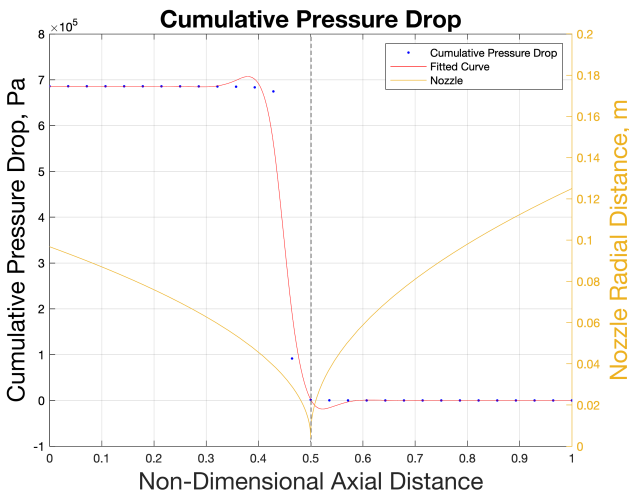


Figure 2.4: Cumulative pressure drop across the nozzle.

2.3 Axial Variations and Power Extraction

So as to quantify the power that can be extracted from the flow to drive the cycle's turbine, a closer look at heat energy transfer must be taken. Diagram 2.5 shows the variation of temperatures from the inside of the nozzle to the coolant channels. The adiabatic wall temperature (T_{aw}) must be decreased within the gas boundary layer to the gas side wall temperature (T_{wg}), i.e. the maximum working temperature of the nozzle wall.

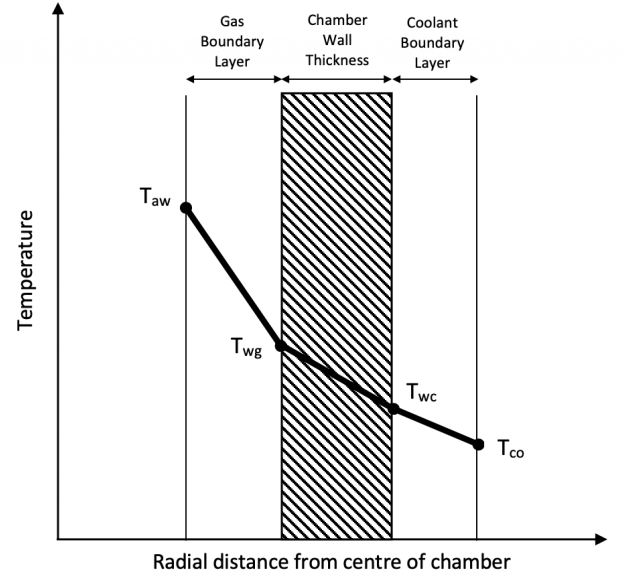


Figure 2.5: Diagram showing the heat transfer from the inside of the nozzle to the coolant channels, created by the author.

The heat flux between layers where the temperature drop can be modelled linearly is calculated using expression 2.3.

$$q = h(T_2 - T_1) \quad (2.3)$$

where h is the heat transfer coefficient: h_g for the gas side boundary layer, k/t for the chamber wall and h_c for the coolant side boundary layer.

The adiabatic wall temperature (T_{aw}) can be calculated from the combustion temperature at every section using equation 2.4, where the local recovery factor (r) for turbulent flows is given by $Pr^{2/3}$ [15].

$$T_{aw} = T_c \left[\frac{1 + r \left(\frac{\gamma-1}{2} \right) M^2}{1 + \left(\frac{\gamma-1}{2} \right) M^2} \right] \quad (2.4)$$

Figure 2.6 illustrates the variation of the adiabatic wall temperature from the nozzle inlet to the nozzle outlet.

NASA's Chemical Equilibrium with Applications (CEA) [18] was utilised to compute the combustion temperature (T_c), ratio of specific heats (γ) and local mach number (M) for each of the 26 sections along the nozzle.

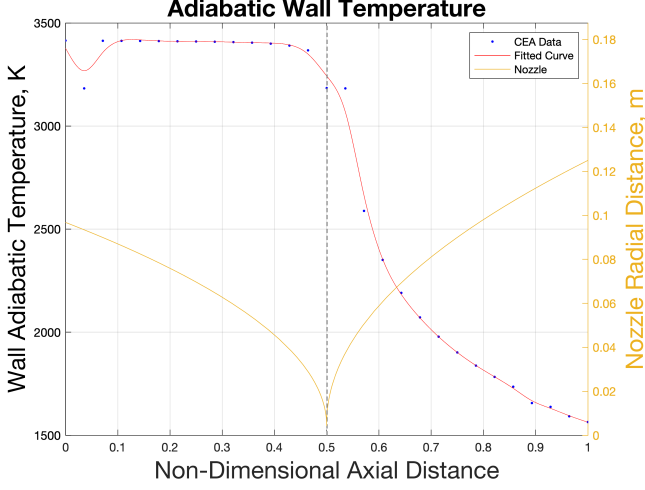


Figure 2.6: Axial variation of adiabatic wall temperature.

The next step consisted in calculating the heat transfer coefficients. The flow of coolants across the smooth channels of the regenerative cooling system is expected to have Reynold's number between 3×10^3 and 10^6 , for which the Dittus Boelter Equation can be used in terms of the Nusselts, Prandtl and Reynolds numbers of the flow [19].

$$Nu = 0.023Re^{0.8}Pr^{0.33} \quad (2.5)$$

Such equation has been modified throughout the years to make it applicable to multiple scenarios. Bartz derived an empirical version for the gas side nozzle heat transfer [20] using the respective expressions for the Nusselt and Reynold's numbers.

$$h_g = \left[\frac{0.026}{D_t^{0.2}} \left(\frac{\mu^{0.2} C_p}{Pr^{0.6}} \right) \left(\frac{pg}{c^*} \right)^{0.8} \left(\frac{A_t}{A} \right)^{0.9} \right] \sigma \quad (2.6)$$

All symbols have their usual meaning and the correction factor, σ , is given by equation 2.7:

$$\sigma = \frac{\left[\frac{1}{2} \frac{T_{wg}}{T_c} \left(1 + \frac{\gamma+1}{2} M^2 \right) + \frac{1}{2} \right]^{-0.68}}{\left[1 + \frac{\gamma+1}{2} M^2 \right]^{0.12}} \quad (2.7)$$

Figure 2.7 shows h_g plotted against the non-dimensionalised nozzle axial distance. As expected, the throat gave the highest values for h_g due to the small area, high energy conditions and the multidimensional nature of the flow [19].

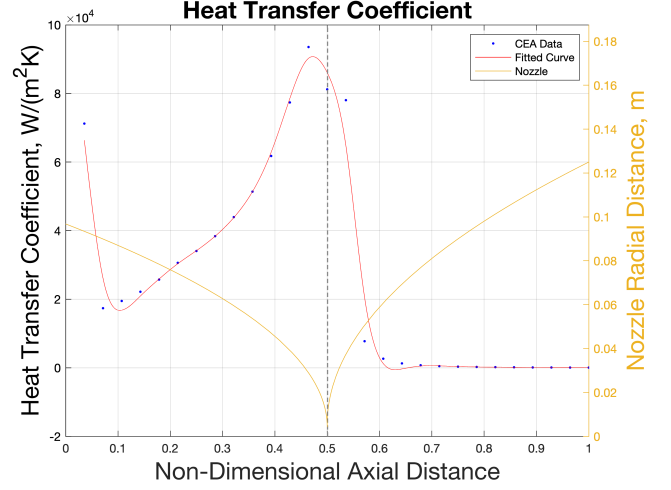


Figure 2.7: Axial variation of heat transfer coefficient.

By taking the maximum working temperature of the nozzle material (T_{wg}) and using equation 2.3, it is possible to determine the heat flow in the gas side boundary layer. Figure 2.8 shows the heat flux across the nozzle.

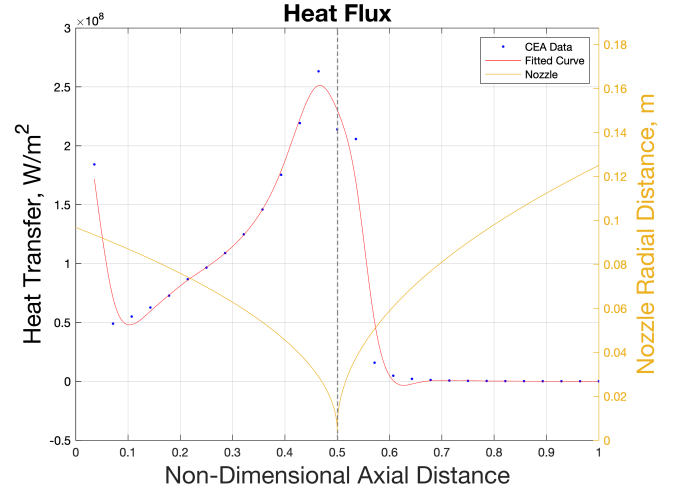
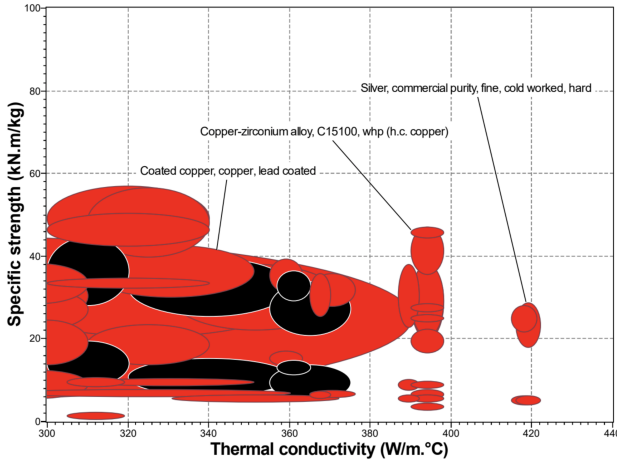


Figure 2.8: Axial variation of the total heat transferred.

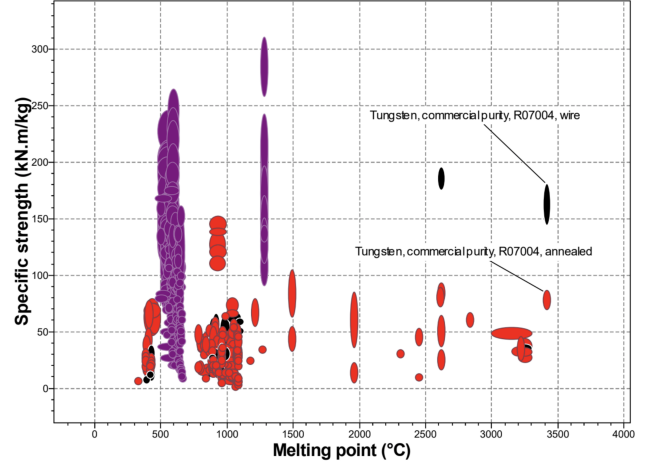
To obtain the maximum power that can be extracted from the cooling system, the surface area of each one of the 26 sections of the nozzle was calculated, assuming cylindrical shape, and multiplied by the local heat flux (q) to obtain the rate of energy transferred to the walls. The total power obtained from one lower stage engine was 279.1126 kW.

Ramana Sabapathy's report contains the calculations which conclude that the turbine only requires 124 kW to drive the pumps. This signifies that this system will tolerate a 44% loss in power between the channels of the cooling system and the inlet of the turbine.

A similar analysis was carried out for the upper stage engine leading to much higher power extraction due to



(a) Ashby chart to identify materials with a high thermal conductivity.



(b) Ashby chart to identify materials with a high melting point.

Figure 2.9: Ashby charts obtained from CES EduPack [1].

its increased surface area. Nevertheless, in order extract the same amount of heat energy as the lower stage engines and ensure the turbine can run smoothly, the regenerative system was shortened to cover the subsonic region of the nozzle and the region just after the throat where heat flux is still extreme.

2.4 Manufacturability

Due to the complexity of its geometry, the cooling jacket will be manufactured additively. Although this method is ideal for complex structures, requires very little skills and has short lead times, the complications in bonding meant that metal alloys, instead of composites, had to be considered for the cooling system [21]. Another disadvantage of additive manufacturing lies in the limitations of bulk production. This was proven to be of little importance due to the system's reusability and minimal requirements for fast manufacturing.

An inexpensive alloy with a high melting point and thermal conductivity would be ideal for this application. A compromise was made between copper-zirconium and tungsten R07004. The former, with a thermal conductivity of $350 \text{ W/m}^\circ\text{C}$ but low melting point 1085°C (see figure 2.9a), will be utilised for the section of the nozzle where heat transfer is most critical: 0 to 0.625 non-dimensional axial distance of figure 2.8. The latter, with a melting point of 3422°C and a poorer thermal conductivity of $115 \text{ W/m}^\circ\text{C}$ (see figure 2.9b), will be used for the remainder of the nozzle, where higher working temperatures are required due to the high aerodynamic drag the engines will suffer when the regenerative cooling system is not operating, during atmospheric descent [1].

Table 2.1: Prices obtained from CES EduPack [1].

Alloy	Price (GBP/kg)
Tungsten R07004	40.40 - 50.70
Copper-Zirconium	4.57 - 5.54
Silver	384.00 - 440.00

Although silver has a higher thermal conductivity than copper, $400 \text{ W/m}^\circ\text{C}$, its expensive price lead to selecting copper instead. See table 2.1 for a price comparison.

2.5 Design of Coolant Channels

The first step towards the design of the regenerative cooling system was quantify the heat flowing into the coolant channels. By employing the known temperature of the flowing coolants and by assuming that the heat transfer from the inside of the nozzle is equal to the heat transfer into the coolant channels, it is possible to calculate the coolant side heat transfer coefficient h_c using equation 2.8.

$$h_c = \frac{q}{T_{co} - T_{cw}} \quad (2.8)$$

Bartz [20] utilised equation 2.5 to derive an additional alternative version for the heat transfer coefficient for propellants at supercritical pressure, h_c , where d is the coolant channel diameter and G the coolant mass per unit area.

$$h_c = \frac{0.029 C_p \mu^{0.2}}{\text{Pr}^{2/3}} \left(\frac{G^{0.8}}{d^{0.2}} \right) \left(\frac{T_{co}}{T_{wc}} \right)^{0.55} \quad (2.9)$$

Therefore, by equating both equations, 2.8 and 2.9, expressing G in terms of the mass flow and the area of the ducts, and using equation 2.10, it is possible to obtain

the optimum values for the number of coolant channels (N), and their diameter (d). Note the 0.8 factor to account for the fact that the tubes are arranged circularly rather than linearly [6].

$$N = \frac{\pi [D_e + 0.8(d + 2t_w)]}{d + 2t_w} \quad (2.10)$$

Figures 2.10 and 2.11 display how the number of channels and diameter change depending on the driving potential, i.e. the difference between the combustion temperature (T_c) and the gas side wall temperature (T_{wg}). The latter based on the limits the nozzle material can be brought to.

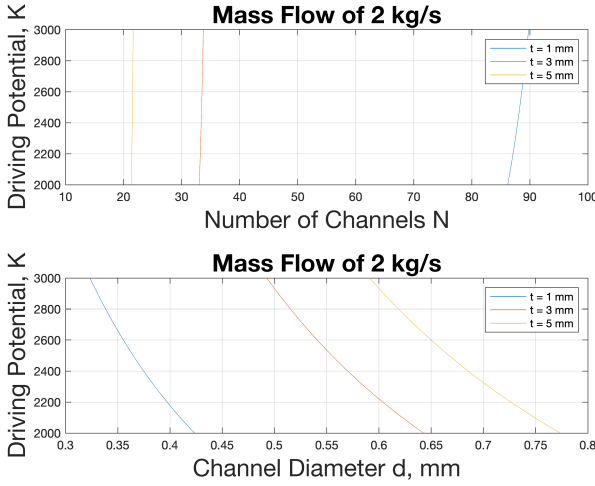


Figure 2.10: Channel design for a constant mass flow of 2 kg/s.

the values of N and d obtained for the throat. N will be kept constant throughout the nozzle, but d will increase depending on the local nozzle diameter.

Figure 2.10 was obtained for an approximate fuel mass flow of 2 kg/s. Tungsten's driving potential can be as low as 725 K, whereas copper's can be as high as 2900 K. For throat conditions, a driving potential of 2500 K was considered for the design of the cooling jacket. A diameter range of 0.35 to 0.65 mm might seem considerably small to the reader. Nevertheless, a value of such order of magnitude would increase the velocity of the fuel flowing through the coolants, optimizing the waste heat extraction. For precision in the manufacturing of the ducts, a wall thickness of 3 mm was chosen for the cooling jacket wall.

Figure 2.11 shows the various possibilities for N and d for a thickness of 3 mm. After finalising the fuel mass flow, 2.39 kg/s, with the Systems Sub-Team, the results in table 2.2 were obtained.

Table 2.2: Final values for the design of the regenerative cooling system.

Driving Potential, K	2500
Number of channels	34
Channel diameter, mm	0.55
Wall Thickness, mm	3.00
Total mass flow, kg/s	2.39

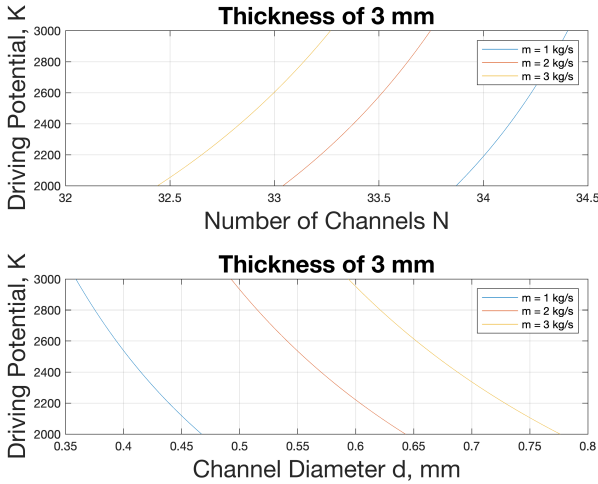


Figure 2.11: Channel design for a constant thickness of 3 mm.

Since the throat is the location where the conditions are the most stringent (see figure 2.8), the cooling system design along the length of the nozzle will be based on

Chapter 3

Fluid Flow Control and Pressurisation

3.1 Rocket Lines

As part of the preliminary and detailed designs of the rocket, it is necessary to size the pipes through which the propellants and pressurants will flow during the mission profile. To do so, a regression data analysis was carried out using the statistics from the NASA's *Liquid Rocket Lines* [2]. Saturn V's F-1, J-2 and SII Engines, the Space Shuttle Main Engine, the Space Shuttle Orbiter Engine and the Titan II Engine were some of the motors considered in this analysis. In order to ensure compatibility between NASA's systems and this lower-scale project, the dimensions obtained were scaled down accordingly depending on the thrust capabilities and rocket dimensions of the engine being considered.

Figures 3.1 and 3.2 show the bellow and flexible hose diameters used for several components within the system. Wide-diameter bellows are used to interconnect the lines and provide integration with other components. Their spring-like shape permits the deflections of the pipes during operation, reduces stresses in the materials and avoids buckling [15]. Bellows will be used at the inlets and outlets of both the turbine and the pumps: these will have a diameter of 70.88 mm. Similarly, the pressurisation system and the hydraulics within the engine will contain bellows of diameters 33.71 mm and 11.19 mm respectively.

Regarding the pipelines, the section joining the pumps to the cooling system will have a diameter of 53.47 mm, whereas the section joining the outlet of the cooling system encompassing all the bypass mechanism to the combustion chamber will have a diameter of 17.36 mm. The lines connecting the pressurant tank to the propellant tanks will have a diameter of 18.34 mm. See figures 3.5 and 3.10 for the location of the pipes.

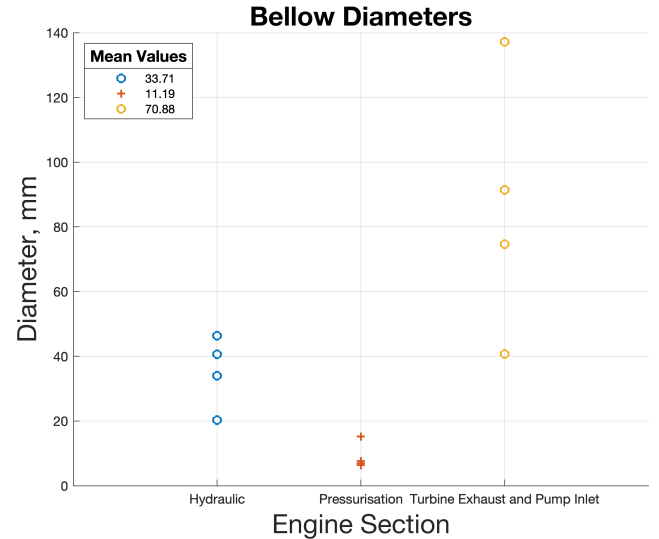


Figure 3.1: Bellow dimensions from regression data, scaled down taking into account thrust capabilities and engine dimensions.

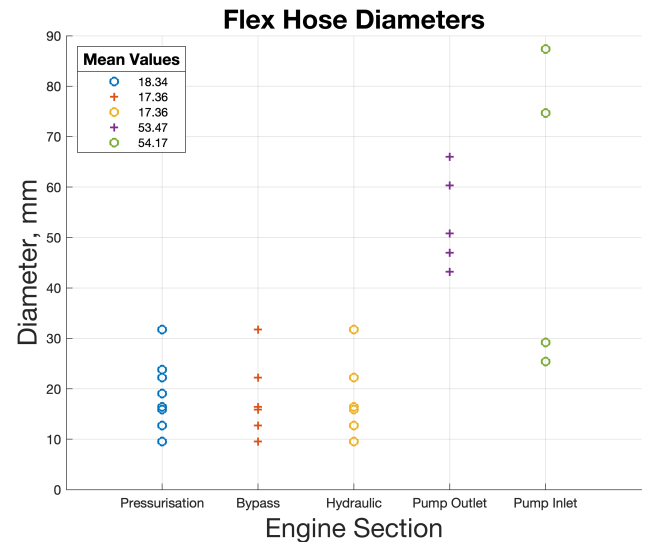


Figure 3.2: Flexible hose dimensions from regression data, scaled down taking into account thrust capabilities and engine dimensions.

Table 3.1 shows a summary of all diameters.

Table 3.1: Selected diameters for *Astraea*'s pipelines.

Type	Section	Mean (mm)
Bellows	Hydraulics	33.71
	Pressurisation	11.19
	Turbine and Pumps	70.88
Flex Hoses	Pressurisation	18.34
	Bypass	17.36
	Hydraulics	17.36
	Pump Outlet	53.47
	Pump Inlet	54.17

3.2 Material Choices for Lines

Pipes and bellows will have to operate at temperatures as low as 90 K, meaning that materials with a high toughness and an invulnerability to cryogenic embrittlement are required. The nickel-based alloy Inconel 718 will be used for the lines joining the pumps to the regenerative cooling system and the turbine, where a high strength-to-weight ratio is required to withstand the 12 MPa pressure outlet of the pumps. After the 5.67 MPa pressure drop across the cooling jacket and turbine, small diameter flex hoses made of 321 CRES (stainless steel) will be employed. See figure 3.5 for pressure drops across the system. Further details on the pressure differences across the pumps and the turbine can be found in Ramana Sabapathy's report.

Both 321 CRES and Inconel 718 can operate at cryogenic temperatures, yet the latter is 59% more expensive than the former due to its higher yield strength and lower thermal expansion coefficient making it more suitable to pipes working under more extreme conditions. Both materials have been widely used in the aerospace industry [2].

Table 3.2: Material properties for 321 CRES and Inconel 718. *Cost relative to 6 ft x 3 in of 6061-T6, data acquired from [2],[3] and [4].

Properties at 300K	321 CRES	Inconel 718
Yield Strength (MPa)	205	550
Thermal Expansion (10^{-6} cm/cm/C)	16.6	13
Cost*	1.5	3.7

3.3 Choice of Valves

Every rocket engine is carefully monitored by a controller. This serves as the brain of the propulsive system. It can provide precise solutions for a full range of scenarios which can occur within the engine. The actuators are equivalent to human muscles. They receive a signal from the controller and transform it into mechanical energy, which is used by the bones of the system, the valves, to elicit movement. This section will explore the types of valves required and their position within the engine (see figure 3.5).

The main oxidizer valve (MOV) and the main fuel valve (MFV) will control the flow of propellants into the cycle [22]. In order to allow for an even heat extraction from the cooling jacket, the flow must limit its turbulence through the regenerative channels. A ball valve is thought to be optimum for this application. Not only is it common for cryogenic propellants but also

provides a quick opening together with little resistance to the flow leading to a small pressure drop [23]. This is highly desirable to ensure that the fuel is still flowing at supercritical pressure. The valve actuator will provide a torque which will rotate a ball with an opening as it can be seen in the figure 3.3.

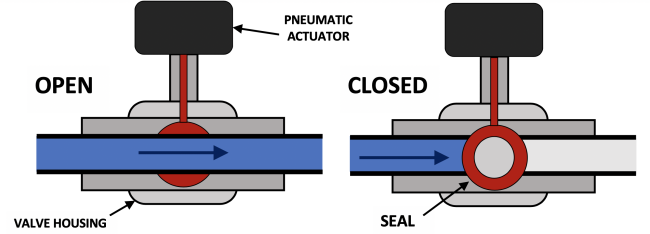


Figure 3.3: Ball valve design, created by the author.

The sizing of the ball valves was carried out using the flow coefficient (see equation 3.1) where SG is the specific gravity of the propellant, Q the flow rate in m^3/h and ΔP the pressure drop in bar.

$$K_v = Q \sqrt{\frac{SG}{\Delta P}} \quad (3.1)$$

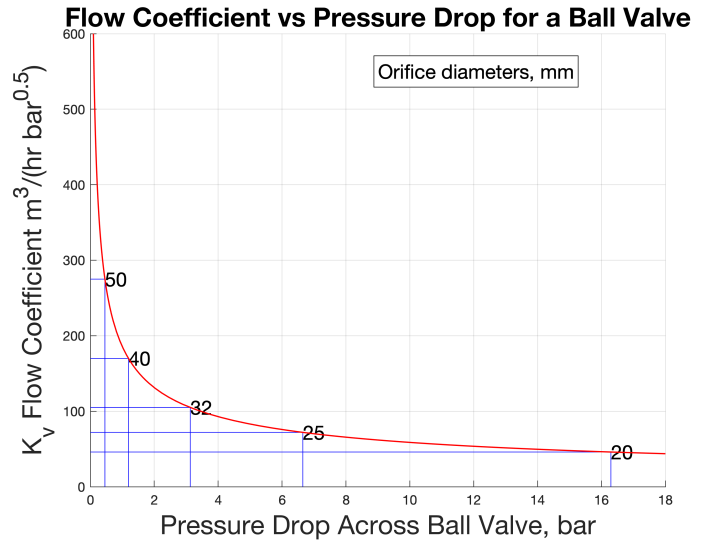


Figure 3.4: Flow coefficient variations with pressure drop for a ball valve.

The *Christian Bürket Manual for Pneumatic Actuator Ball Valves* [24], was utilised to obtain values of K_v for different orifice diameters, as shown in figure 3.4. An orifice of 25 mm was selected to ensure compatibility with the pipelines from the pumps to the cooling system, leading to a pressure loss of 6.545 bar across the MFV and the MOV. See figure 3.5 for the pressure drops across the system.

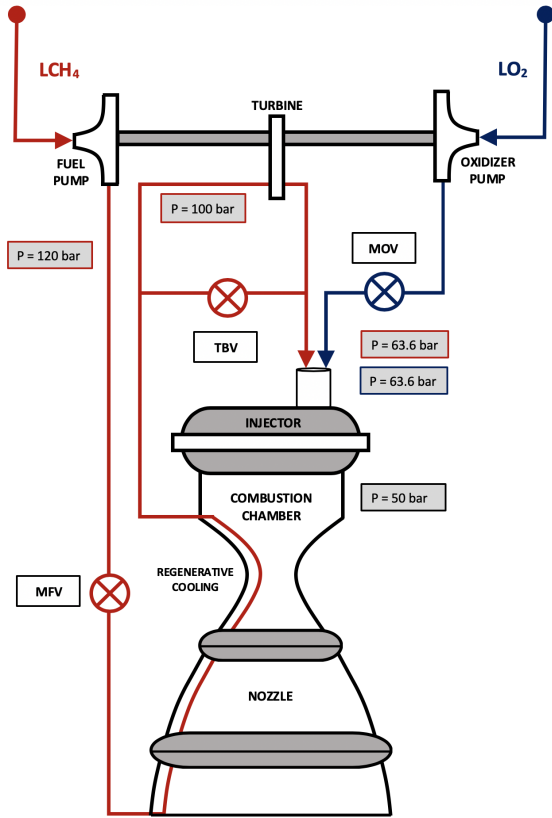


Figure 3.5: Schematic layout of *Mantis* showing the pressure drops across the system, created by the author.

The turbine bypass valve (TBV) will be a globe valve and will serve as a pressure controller [23]. Temperature is inversely proportional to density. Thus, were there to be an ignition instability in the combustion chamber, the increased heat absorbed by the cooling system would lead to lower pressure losses across the cooling jacket (see equation 2.1). This would influence the pressure conditions downstream of the turbine, leading to further combustion instabilities. To avoid this, the TBV would open to allow some flow and reduce the pressure in the system, ensuring that the pumps are also running smoothly at the expected power.

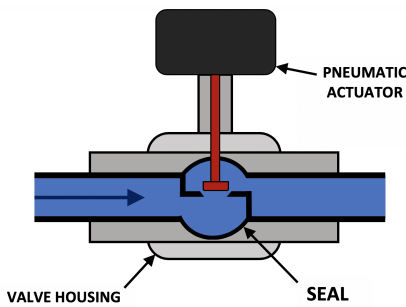


Figure 3.6: Globe valve design, created by the author.

Globe valves have very large pressure drops. As it can

be seen in figure 3.6, the very nature of the mechanism considerably obstructs the flow. This could be used as an advantage. Since the bypass valve will be located in a parallel configuration to the turbine, ideally it should provide the same pressure drop as the turbine. Throttling can be achieved by opening the globe valve and increasing the flow of fuel to the combustion chamber.

The sizing of the globe valve was carried out in a similar manner to the sizing of the ball valve. The *Christian Bürket Manual for Pneumatic Actuator Globe Valves* [25] was used, yet this time the aim was to replicate the pressure drop across the turbine, 36.4 bar. The orifice diameter chosen was 45 mm.

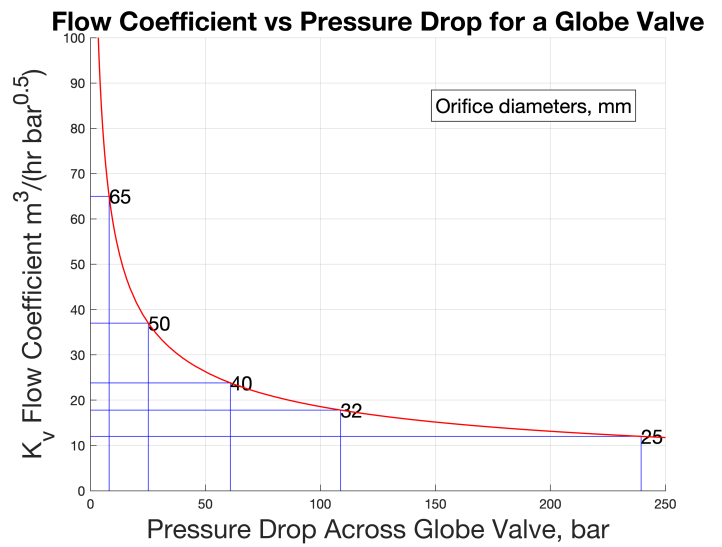


Figure 3.7: Flow coefficient variations with pressure drop for a globe valve.

Additional fluid flow control devices such as pressure relief valves and check valves will be positioned throughout the system to ensure the pressure of the flow and its direction behave as expected [26].

Expander cycle rocket actuators are generally electric or pneumatic (gas-driven) [10]. Even though electric actuators were deemed appropriate for rocket engine applications due to their high level of precision, they are unable to return to a fail-safe position in case of failure and are not guaranteed to work satisfactorily under the system's cryogenic working environment. In addition to this, their instantaneous response to signals from the controller increases the chances of shock waves occurring in the pipes leading to damaged engine components [23].

Low-cost and reusability are some of the main drivers of this project, so to reduce financial pressure from the piping system and guarantee that the main engine parts can still be utilised after failure, pneumatically actuated

valves were chosen instead.

To achieve full control of the flow, a control loop is employed around the valves (see figure 3.8). A sensor will capture the flow parameters. These are then analysed by a flow controller and compared to set values. If there is a significant difference, the flow controller will send an electric signal into the I/P transducer which will output a pneumatic signal into the actuator to rectify the control valve's behaviour [23].

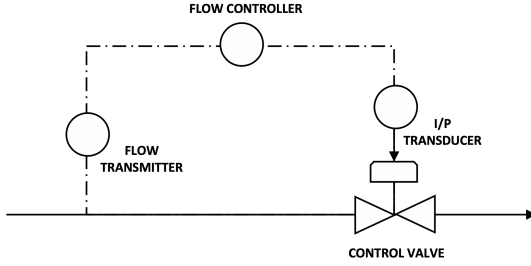


Figure 3.8: Flow control loop, created by the author.

3.4 Pressurisation and Helium Mass Requirements

From a structural point of view, the propellant tanks must be fully pressurised during descent to ensure buckling does not take place. This involves injecting a high-pressure gas into the ullage of the tanks, as seen in figure 3.9. A trade-off analysis was carried out to determine the best medium with which the pressurisation of the tanks can be carried out [27].

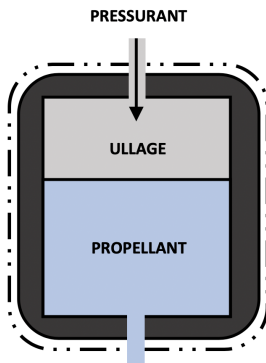


Figure 3.9: Ullage inside propellant tanks, created by the author.

The product of the combustion gases has been employed by the Titan vehicles for the pressurisation system [27]. This configuration would offer a lighter architecture since an additional tank containing the pressurant gas would not be required. Unfortunately,

the compatibility of the combustion gases would only be adequate with the fuel propellant. In addition to this, the high temperature of the exhaust would further complicate its employment and the reusability of the entire pressurisation system would not be feasible.

Another option would be to utilise an evaporated version of the propellants. This method is more feasible thanks to the regenerative cooling system which could heat up the propellants. By reducing the pressure of the fuel using a pressure relief valve, the methane could transition from its supercritical phase into a gaseous one for pressurisation. Nevertheless, this option was ruled out due to the large ullage pressure decay, which can arise from the condensation of the gas at the interface with the liquid, and the lack of compatibility with the oxidizer [27].

The results of the trade-off analysis demonstrated that employing another inert gas is the best option. Although the need to have an additional tank to accommodate the gas might not be seen as structurally optimal, this option was chosen due to its compatibility with both the propellants and its higher reliability. Helium was selected due to its low weight.

Figure 3.10 shows a simplified version of a layout proposal for the pressurisation system. This configuration would involve the use of a throttling globe valve for start-up and shut-off (SV). The pressure drop across the valve would be compensated by the use of a heat exchanger which would heat up the helium, increasing its pressure and volume and thus lowering the mass required to pressurise the tanks. A pressure regulator consisting of a controller operating a regulator valve would monitor the pressure before the propellant tanks. This system's approach is similar to that of the control valves, in that an I/P transducer would also be in charge of any corrections between the output and input pressures.

The next step was to establish the mass of the pressurant required. This was done by finding the heat transfer from the liquid propellant to the ullage gas using 3.2.

$$Q = H A t (T_u - T_e) \quad (3.2)$$

H is an experimentally determined heat transfer coefficient for a gas-liquid interface, whose value, $1.6299 \text{ J/m}^2\text{Ks}$, was obtained from *Modern Design of Liquid Propellant Engines* [15]. T_u and T_e are the ullage gas and propellant liquid temperatures respectively.

The value for the heat energy flux (Q) is assumed to be equal to that which vaporized the propellant liquid into ullage gaseous state, calculated using C_{p_l} , C_{v_l} , h_v : the liquid and vaporised specific heat capacities of the propellant and its latent heat respectively.

$$Q = M_v [C_{p_l}(T_v - T_e) + h_v + C_{p_v}(T_u - T_v)] \quad (3.3)$$

Therefore, by equating both expressions and rearranging to find the M_v , it is possible to obtain the mass of the vaporised propellant. Consequently, by employing the ideal gas equation, where R is the specific gas constant, the volume V_v of the vaporised propellant was determined.

$$pV = MRT \quad (3.4)$$

The volume of pressurant gas contained in the tank was then found using $V_{He} = V_T - V_v$. Subsequently, by using equation 3.4 again, the required mass of helium to keep every tank pressurised to 3 bar, the optimum pressure provided by the Structures Sub-Team, was quantified. Table 3.3 includes the results of the calculation. The same analysis was carried out using nitrogen instead, yet its much lower specific gas constant meant that the total weight required would be 103.7612 kg.

Table 3.3: Helium mass required, all values in kilograms.

	Lower Stage	Upper Stage
LOX tank (kg)	6.4963	1.2374
LCH4 tank (kg)	5.9561	1.1375
Total (kg)	14.8273	

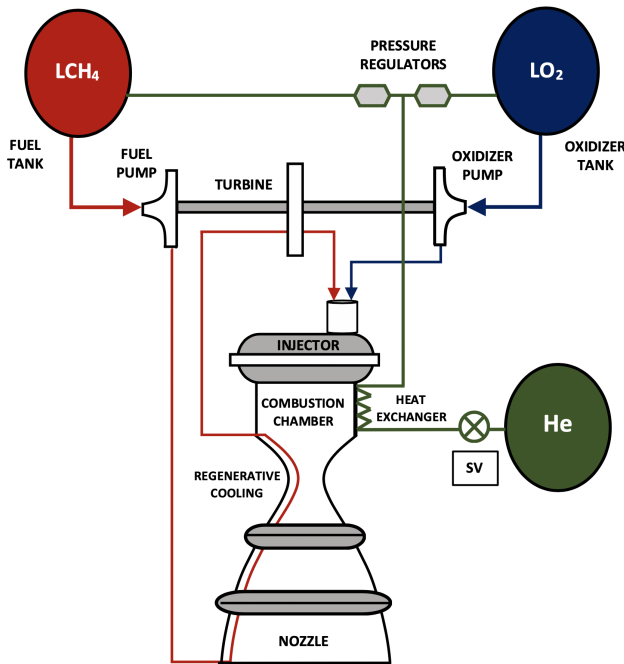


Figure 3.10: Schematic layout of pressurisation system, created by the author.

3.5 Discussion

Mantis, the propulsive unit that will serve *Astraea*, will be a LOX/LCH4 expander cycle liquid bi-propellant engine. This report has proved that it is possible to develop a lower stage expander cycle, since the power extracted from waste heat at the nozzle and the combustion chamber, 279.1126 kW, is well over the turbine requirements to power the pumps, 124 kW. The regenerative cooling system that will render this possible will have 34 distinct ducts through which the fuel will absorb the nozzle's waste heat. The pressure drop across the cooling jacket will be 6.8584 bar.

So as to ease the production of *Mantis*, additive manufacturing will be implemented. The nozzle will make use of high thermal conductivity and high melting point materials, such as copper-zirconium and tungsten R07004 respectively. The piping system, however, will require materials with a high strength-to-weight ratio, so Inconel 718 and 321 CRES will be used instead.

For compatibility purposes and to avoid ullage decay, an inert gas will pressurise the propellant tanks. Helium was selected due to its low weight and the mass required, 14.8273 kg, was calculated.

Fluid flow control devices will be placed throughout the system. The main fuel and oxidizer valves must provide a tight seal yet must not induce a high pressure drop in the flow. Ball valves, which have been known to work excellently under cryogenic conditions, are optimal for this. The turbine bypass system will operate with a globe valve, which will provide the required pressure drop and throttability.

Reusability, the main driver of the project, has been taken into account throughout the design of *Mantis*. Soot deposition has been minimized by choosing methane as the fuel and by selecting an expander cycle instead of other more common systems, such as gas-generator pre-burners, which require reparation and the dismantling of the engine for soot cleaning. Mechanisms such as fail-safe pneumatic actuator valves will be put in place to avoid component damage in case of failure.

As a matter for further developments, computational fluid mechanics could be made use of to properly model the flow of propellants through the system. This would enable a more accurate calculation of the pressure drops across the expander cycle together with a more precise value for the cooling system power extraction.

A model on SIMULINK™ could also be developed. A series of closed-loop feedback models to control and operate the valves could be put in place to ensure that the flow can be controlled and the conditions at the combustion chamber are optimised for ignition.

Appendix A

Raw Data

A.1 CEA Lower Stage

LOWER

mach	t	gam	cp	p	ae
0.0000E+00	3.4145E+03	1.1399E+00	5.8652E+00	5.0000E+01	0.0000E+00
1.0000E+00	3.2267E+03	1.1405E+00	5.3156E+00	2.8814E+01	1.0000E+00
3.3227E+00	1.8448E+03	1.2079E+00	2.2990E+00	5.0000E-01	1.2534E+01
6.6409E-02	3.4137E+03	1.1399E+00	5.8629E+00	4.9875E+01	9.0000E+00
7.4759E-02	3.4134E+03	1.1399E+00	5.8623E+00	4.9841E+01	8.0000E+00
8.5517E-02	3.4131E+03	1.1399E+00	5.8614E+00	4.9792E+01	7.0000E+00
9.9909E-02	3.4126E+03	1.1398E+00	5.8600E+00	4.9716E+01	6.0000E+00
1.2018E-01	3.4117E+03	1.1398E+00	5.8578E+00	4.9590E+01	5.0000E+00
1.3378E-01	3.4110E+03	1.1398E+00	5.8560E+00	4.9493E+01	4.5000E+00
1.5090E-01	3.4100E+03	1.1398E+00	5.8535E+00	4.9356E+01	4.0000E+00
1.7312E-01	3.4086E+03	1.1398E+00	5.8498E+00	4.9154E+01	3.5000E+00
2.0319E-01	3.4064E+03	1.1398E+00	5.8440E+00	4.8839E+01	3.0000E+00
2.4637E-01	3.4026E+03	1.1398E+00	5.8339E+00	4.8303E+01	2.5000E+00
3.1427E-01	3.3951E+03	1.1398E+00	5.8142E+00	4.7273E+01	2.0000E+00
4.4076E-01	3.3766E+03	1.1398E+00	5.7641E+00	4.4793E+01	1.5000E+00
9.9655E-01	3.2280E+03	1.1405E+00	5.3196E+00	2.8921E+01	1.0000E+00
1.0000E+00	3.2267E+03	1.1405E+00	5.3156E+00	2.8814E+01	1.0000E+00
2.0110E+00	2.7259E+03	1.1576E+00	3.5835E+00	6.4624E+00	2.0000E+00
2.3279E+00	2.5177E+03	1.1714E+00	3.0286E+00	3.5463E+00	3.0000E+00
2.5344E+00	2.3738E+03	1.1815E+00	2.7539E+00	2.3594E+00	4.0000E+00
2.6898E+00	2.2642E+03	1.1886E+00	2.5989E+00	1.7307E+00	5.0000E+00
2.8155E+00	2.1764E+03	1.1938E+00	2.5031E+00	1.3475E+00	6.0000E+00
2.9213E+00	2.1037E+03	1.1977E+00	2.4398E+00	1.0924E+00	7.0000E+00
3.0130E+00	2.0421E+03	1.2006E+00	2.3956E+00	9.1189E-01	8.0000E+00
3.0940E+00	1.9887E+03	1.2029E+00	2.3636E+00	7.7819E-01	9.0000E+00
3.1665E+00	1.9420E+03	1.2047E+00	2.3394E+00	6.7568E-01	1.0000E+01
3.2925E+00	1.8632E+03	1.2074E+00	2.3058E+00	5.2980E-01	1.2000E+01
3.3995E+00	1.7988E+03	1.2093E+00	2.2838E+00	4.3180E-01	1.4000E+01
3.4475E+00	1.7706E+03	1.2100E+00	2.2755E+00	3.9414E-01	1.5000E+01

A.2 CEA Upper Stage

UPPER

mach	t	gam	cp	p	ae
0.0000E+00	3.4145E+03	1.1399E+00	5.8652E+00	5.0000E+01	0.0000E+00
1.0000E+00	3.2267E+03	1.1405E+00	5.3156E+00	2.8814E+01	1.0000E+00
4.9158E+00	1.1197E+03	1.2138E+00	2.2344E+00	2.9412E-02	1.1192E+02
6.6409E-02	3.4137E+03	1.1399E+00	5.8629E+00	4.9875E+01	9.0000E+00
7.4759E-02	3.4134E+03	1.1399E+00	5.8623E+00	4.9841E+01	8.0000E+00
8.5517E-02	3.4131E+03	1.1399E+00	5.8614E+00	4.9792E+01	7.0000E+00
9.9909E-02	3.4126E+03	1.1398E+00	5.8600E+00	4.9716E+01	6.0000E+00
1.2018E-01	3.4117E+03	1.1398E+00	5.8578E+00	4.9590E+01	5.0000E+00
1.3378E-01	3.4110E+03	1.1398E+00	5.8560E+00	4.9493E+01	4.5000E+00
1.5090E-01	3.4100E+03	1.1398E+00	5.8535E+00	4.9356E+01	4.0000E+00
1.7312E-01	3.4086E+03	1.1398E+00	5.8498E+00	4.9154E+01	3.5000E+00
2.0319E-01	3.4064E+03	1.1398E+00	5.8440E+00	4.8839E+01	3.0000E+00
2.4637E-01	3.4026E+03	1.1398E+00	5.8339E+00	4.8303E+01	2.5000E+00
3.1427E-01	3.3951E+03	1.1398E+00	5.8142E+00	4.7273E+01	2.0000E+00
4.4076E-01	3.3766E+03	1.1398E+00	5.7641E+00	4.4793E+01	1.5000E+00
9.9655E-01	3.2280E+03	1.1405E+00	5.3196E+00	2.8921E+01	1.0000E+00
1.0000E+00	3.2267E+03	1.1405E+00	5.3156E+00	2.8814E+01	1.0000E+00
2.6898E+00	2.2642E+03	1.1886E+00	2.5989E+00	1.7306E+00	5.0000E+00
3.1665E+00	1.9420E+03	1.2047E+00	2.3394E+00	6.7568E-01	1.0000E+01
3.6491E+00	1.6575E+03	1.2126E+00	2.2486E+00	2.6992E-01	2.0000E+01
3.9371E+00	1.5101E+03	1.2149E+00	2.2262E+00	1.5903E-01	3.0000E+01
4.1449E+00	1.4136E+03	1.2158E+00	2.2183E+00	1.0958E-01	4.0000E+01
4.3083E+00	1.3433E+03	1.2160E+00	2.2161E+00	8.2206E-02	5.0000E+01
4.4433E+00	1.2886E+03	1.2159E+00	2.2168E+00	6.5060E-02	6.0000E+01
4.5586E+00	1.2443E+03	1.2156E+00	2.2190E+00	5.3418E-02	7.0000E+01
4.6593E+00	1.2073E+03	1.2153E+00	2.2221E+00	4.5053E-02	8.0000E+01
4.7488E+00	1.1757E+03	1.2149E+00	2.2258E+00	3.8781E-02	9.0000E+01
4.8293E+00	1.1482E+03	1.2144E+00	2.2297E+00	3.3924E-02	1.0000E+02
4.9368E+00	1.1129E+03	1.2137E+00	2.2357E+00	2.8418E-02	1.1500E+02

A.3 Data from NASA's Rocket Lines

Table A.1: Raw data obtained from NASA's *Rocket Lines* [2], used for pipeline sizing. Pressure and temperature conditions omitted.

Rocket		Part Description	Fluid	Line Diameter (in) low	Line Diameter (in) high
Saturn V F-1	Small flex hoses	Tank pressurisation, bleed and bypass functions, hydraulic system	LOX	0.75	1
			RP-1	0.25	2.25
			He	0.25	0.75
			GN2	0.5	0.75
			Exhaust gas	0.5	
	Bellows	Propellant feed, hydraulic system, drain lines	LOX	1.25	17
			RP-1	1.37	12
		Tank pressurisation	GOX	1.25	
			He	1.25	1.5
		Turbine exhaust and pump inlets	Exhaust gas	27	
			LOX	18	
			RP-1	12	17.4
Saturn V J-2	Small flex hoses	Tank pressurisation, bleed and bypass functions, hydraulic system	LOX	0.375	1.5
			GOX	0.5	1.25
			He	0.25	0.75
			LH2	0.25	1.5
			GH2	0.5	2
			LN2	0.375	
			GN2	0.375	
	Bellow	Pump discharge	LOX	4	
			LH2	4	
		Pump inlet	LOX	8	
			LH2	8	
		Turbine exhaust	LO2/LH2	4	12
Space Shuttle Main Engine		Discharge duct, low pressure fuel pump	LH2	5.2	
		Supply duct to turbine for low-pressure fuel pump	Hot gas	2	
		Supply duct to turbine for low-pressure oxidizer pump	Hot gas	2.3	
		Oxidizer tank pressurisation duct	GOX	0.75	
		High-pressure oxidizer pump discharge	LOX	4	
		High-pressure fuel pump discharge	LH2	3.4	
Shuttle Orbiter		Fuel lines, external tank disconnect to manifold	LH2	16.85	
		Oxidizer line to engines	LOX	16.85	
		Oxidizer fill and drain duct	LOX	8	
		External tank pressurisation	GOX	2	
Saturn V SII	Braided hoses	Pressurisation, drain functions	GH2	0.29	1
			He	0.25	1
	Bellows	Propellant feed	LOX	8	
			LH2	8	
		LOX tank pressurisation	GOX	1.5	
			GOX	3	
			GOX	3	
		LH2 Pressurization	GH2	1.5	
			GH2	3	
			GH2	3	
Titan II		Pump inlet	A-50	5.88	
			N2O4	6.88	
		Pump discharge	A-50	3.7	
			N2O4	4.75	
		Tank pressurisation			
			Hot gas	0.75	

Appendix B

MATLAB Code

```
1 %% CODE FOR ENGINE REGENERATIVE COOLING DESIGN
2 % Alvaro Fieira Wachtendorff, 25/05/2020
3
4 %OUTPUT GRAPHS:
5 % CHANNEL DESIGN FOR THROAT
6 % AXIAL HEAT TRANSFER LOWER STAGE
7 % AXIAL HEAT TRANSFER UPPER STAGE
8 % DENSITY VARIATIONS FOR SUPERCRITICAL METHANE
9 % AXIAL AND CUMULATIVE PRESSURE DROP
10 % BALL VALVE FLOW COEFFICIENT
11 % GLOBE VALVE FLOW COEFFICIENT
12 % FLEX HOSE DIAMETERS
13 % BELLOW DIAMETERS
14 % METHANE PHASE DIAGRAM
15
16 %OUTPUT VALUES:
17 % TOTAL POWER EXTRACTION FOR LOWER AND UPPER STAGES
18 % TOTAL PRESSURE DROP ACROSS COOLING SYSTEM
19 % TOTAL HELIUM REQUIRED FOR PRESSURIZATION
20
21 %Housekeeping
22
23 clc
24 clear
25 close all
26
27 %% PARAMETERS
28
29 % INPUTS: T_WG AND T_CO, DRIVING POTENTIAL IS T_OG - T_WG
30
31 %Throat Parameters from CEA
32
33 T_wg = 700; %K, wall temperature
34 M = 1; %throat mach number (choked)
35 gamma = 1.1405; %specific heats from CEA
36 Pr_g = 4*gamma / (9*gamma - 1 ); %Prandtl number of combustion gases
37
38 Dt = 0.0645492; %m, throat diameter
39 Molec_mass_throat = 20.433; %from CEA
40 Temp_throat = 3226.72; %K, from CEA
41 visc_g = Molec_mass_throat^0.5*Temp_throat^0.6b@x...
42     *46.6*10^-10; %SI units, dynamic visc of combustion gases CHECK
```

```

43 Cp_g = 5.3156*10^3; %SI units, specific heat capacity of comb gases
44 Pc = 5 * 10^6; %Pa, combustion chamber pressure
45 g = 9.81; %m/s^2, gravitational acceleration
46 c = 1861.9; %m/s, characteristic velocity from CEA
47 At_A = 1;
48
49 %Wall Parameters
50
51 t = [0.001;0.003;0.005]; %m, wall thickness CHECK
52 k_cond = 400; %SI units, thermal conductivity of wall (from material)
53
54 %Coolant Parameters
55
56 Cp_c = 3.6*10^3; %SI units, specific heat of coolant
57 visc_c = 2.5 * 10^-5; %SI units, viscosity of coolant
58 Pr_c = 0.8; %prandtl number of coolant for methane at -150C
59 m_f = [1,2,3]; %kg/s, total fuel flow rate of coolant
60 T_co = 200; %K, coolant temp at throat ASSUMED
61
62
63 %% EQUATIONS
64
65 %Combustion gases temperature
66 T_og = (2700:1:3700); %if 1200:1800, change fzero to 0.1 and T_wg = 1000;
67
68 %Wall adiabatic temperature
69 T_aw = T_og.* (1 + Pr_g^0.33*M^2*(gamma-1)/2) / (1 + M^2*(gamma-1)/2);
70
71 %Correction factor for gas heat transfer parameter
72 sigma = (0.5.*(T_wg./T_og).*(1+M^2*(gamma-1)/2)+0.5).^(-0.68...
73         .*(1+M^2*(gamma-1)/2).^(-0.12);
74
75 %Bartz gas heat transfer parameter
76 h_g = (0.026/Dt^0.2)*(Cp_g*visc_g^0.2/Pr_g^0.6)...
77         *(Pc*g/c)^0.8*(At_A)^0.9*sigma;
78
79 %Heat transfer from hot gas side
80 q = h_g .* (T_aw - T_wg);
81
82 %Heat transfer within side walls and factor of 1000
83 T_wc = T_wg - t.*q/(k_cond*100);
84
85 %Heat transfer in coolant
86 h_c = q ./ (T_wc - T_co);
87
88 %Bartz equation for supercritical coolant (vapour-film boundary layer)
89 coeff = zeros(3,1001,3);
90 for k = 1:1:3
91     coeff(:, :, k) = ( ( 0.029* Cp_c * visc_c^0.2 ./ (h_c .* Pr_c^(2/3)) ) ...
92         .* ( 4.*m_f(k) ./ pi ).^0.8 .* ( T_co ./ T_wc).^0.55 ).^(1/0.8);
93 end
94
95 N = zeros; %number of channels
96 d = zeros; %channel diameter
97
98 %Work out N and d iteratively using fzero
99

```

```
100 for i = 1:1:length(T_og) %for different combustion gases
101
102     for j = 1:1:length(t) %for different wall thicknesses
103
104         for k = 1:1:length(m_f) %for different total mass flows
105
106             % N = pi * (Dt + 0.8*(d+2*t))/(d+2*t);
107             % N = coeff * d^-2.25;
108
109             func = @(d) coeff(j,i,k) .* d.^(-2.25) ...
110                 - pi * (Dt + 0.8*(d+2*t(j)))/(d+2*t(j));
111
112             d(j,i,k) = fzero(func, 0.001);
113             N(j,i,k) = coeff(j,i,k) * d(j,i,k)^(-2.25);
114
115         end
116
117     end
118
119 end
120
121 %% CHANNEL DESIGN PLOTS
122
123 %sgtitle('REGENERATIVE COOLING DESIGN','FontWeight','bold','FontSize',20);
124
125 figure
126
127 tiledlayout(2,1)
128
129 nexttile
130 plot(N(:, :, 2), T_og-T_wg);
131 ylabel('Driving Potential, K','FontSize',20)
132 xlabel('Number of Channels N', 'FontSize',20);
133 xlim([10 100]);
134 ylim([2000 3000])
135 hold on
136 %plot([0 100],[Temp_throat-T_wg Temp_throat-T_wg])
137 grid on
138 title('Mass Flow of 2 kg/s','FontSize',20);
139 legend('t = 1 mm','t = 3 mm','t = 5 mm');
140
141 nexttile
142 plot(d(:, :, 2)*10^3, T_og-T_wg);
143 ylim([2000 3000])
144 ylabel('Driving Potential, K','FontSize',20)
145 xlabel('Channel Diameter d, mm', 'FontSize',20);
146 grid on
147 title('Mass Flow of 2 kg/s','FontSize',20);
148 legend('t = 1 mm','t = 3 mm','t = 5 mm');
149
150 print('channeldesignmassflow.png','-dpng','-r400');
151
152 figure
153 tiledlayout(2,1)
154
155 nexttile
156 plot(N(2, :, 1), T_og-T_wg);
```

```
157 hold on
158 plot(N(2,:,2), T_og-T_wg);
159 hold on
160 plot(N(2,:,3), T_og-T_wg);
161 hold off
162 ylabel('Driving Potential, K','FontSize',20)
163 xlabel('Number of Channels N', 'FontSize',20);
164 grid on
165 ylim([2000 3000])
166 title('Thickness of 3 mm','FontSize',20);
167 legend('m = 1 kg/s','m = 2 kg/s','m = 3 kg/s');
168
169 nexttile
170 plot(d(2,:,1)*10^3, T_og-T_wg);
171 hold on
172 plot(d(2,:,2)*10^3, T_og-T_wg);
173 hold on
174 plot(d(2,:,3)*10^3, T_og-T_wg);
175 hold off
176 ylabel('Driving Potential, K','FontSize',20)
177 xlabel('Channel Diameter d, mm', 'FontSize',20);
178 grid on
179 ylim([2000 3000])
180 title('Thickness of 3 mm','FontSize',20);
181 legend('m = 1 kg/s','m = 2 kg/s','m = 3 kg/s');
182
183 print('channeldesignthickness.png','-dpng','-r400');
184
185
186 %% CEA DATA
187
188 %%% CEA INPUTS:
189
190 % Chamber pressure = 50 bar
191 % Fuels = LOX/LCH4
192 % Mixing ratio = 2.95
193
194 % Pc/Pe = 50 bar / 0.5 bar for lower stage
195 % Pc/Pe = 50 bar / 2.94kPa for upper stage
196
197 % Ae/At = 15 for lower stage
198 % Ae/At = 115 for upper stage
199 % Ac/At = 9 chamber ratio
200
201 % Area ratios used for lower stage (13 values allowed for each regime) =
202 % Subsonic: 9 8 7 6 5 4.5 4 3.5 3 2.5 2 1.5 1
203 % Supersonic: 1 2 3 4 5 6 7 8 9 10 12 14 15
204
205 % Area ratios used for upper stage (13 values allowed for each regime) =
206 % Subsonic: 9 8 7 6 5 4.5 4 3.5 3 2.5 2 1.5 1
207 % Supersonic: 1 5 10 20 30 40 50 60 70 80 90 100 115
208
209 %%% CEA OUTPUTS:
210
211 %CEA Data for Lower Stage
212 Data_Lower_Stage = ...
213 [0.0000E+00 3.4145E+03 1.1399E+00 5.8652E+00 5.0000E+01 0.0000E+00
```

```
214 1.0000E+00 3.2267E+03 1.1405E+00 5.3156E+00 2.8814E+01 1.0000E+00
215 6.6409E-02 3.4137E+03 1.1399E+00 5.8629E+00 4.9875E+01 9.0000E+00
216 7.4759E-02 3.4134E+03 1.1399E+00 5.8623E+00 4.9841E+01 8.0000E+00
217 8.5517E-02 3.4131E+03 1.1399E+00 5.8614E+00 4.9792E+01 7.0000E+00
218 9.9909E-02 3.4126E+03 1.1398E+00 5.8600E+00 4.9716E+01 6.0000E+00
219 1.2018E-01 3.4117E+03 1.1398E+00 5.8578E+00 4.9590E+01 5.0000E+00
220 1.3378E-01 3.4110E+03 1.1398E+00 5.8560E+00 4.9493E+01 4.5000E+00
221 1.5090E-01 3.4100E+03 1.1398E+00 5.8535E+00 4.9356E+01 4.0000E+00
222 1.7312E-01 3.4086E+03 1.1398E+00 5.8498E+00 4.9154E+01 3.5000E+00
223 2.0319E-01 3.4064E+03 1.1398E+00 5.8440E+00 4.8839E+01 3.0000E+00
224 2.4637E-01 3.4026E+03 1.1398E+00 5.8339E+00 4.8303E+01 2.5000E+00
225 3.1427E-01 3.3951E+03 1.1398E+00 5.8142E+00 4.7273E+01 2.0000E+00
226 4.4076E-01 3.3766E+03 1.1398E+00 5.7641E+00 4.4793E+01 1.5000E+00
227 9.9655E-01 3.2280E+03 1.1405E+00 5.3196E+00 2.8921E+01 1.0000E+00
228 1.0000E+00 3.2267E+03 1.1405E+00 5.3156E+00 2.8814E+01 1.0000E+00
229 2.0110E+00 2.7259E+03 1.1576E+00 3.5835E+00 6.4624E+00 2.0000E+00
230 2.3279E+00 2.5177E+03 1.1714E+00 3.0286E+00 3.5463E+00 3.0000E+00
231 2.5344E+00 2.3738E+03 1.1815E+00 2.7539E+00 2.3594E+00 4.0000E+00
232 2.6898E+00 2.2642E+03 1.1886E+00 2.5989E+00 1.7307E+00 5.0000E+00
233 2.8155E+00 2.1764E+03 1.1938E+00 2.5031E+00 1.3475E+00 6.0000E+00
234 2.9213E+00 2.1037E+03 1.1977E+00 2.4398E+00 1.0924E+00 7.0000E+00
235 3.0130E+00 2.0421E+03 1.2006E+00 2.3956E+00 9.1189E-01 8.0000E+00
236 3.0940E+00 1.9887E+03 1.2029E+00 2.3636E+00 7.7819E-01 9.0000E+00
237 3.1665E+00 1.9420E+03 1.2047E+00 2.3394E+00 6.7568E-01 1.0000E+01
238 3.2925E+00 1.8632E+03 1.2074E+00 2.3058E+00 5.2980E-01 1.2000E+01
239 3.3227E+00 1.8448E+03 1.2079E+00 2.2990E+00 5.0000E-01 1.2534E+01
240 3.3995E+00 1.7988E+03 1.2093E+00 2.2838E+00 4.3180E-01 1.4000E+01
241 3.4475E+00 1.7706E+03 1.2100E+00 2.2755E+00 3.9414E-01 1.5000E+01];
242
243 %CEA Data for Upper Stage
244 Data_Upper_Stage = [...
245 0.0000E+00 3.4145E+03 1.1399E+00 5.8652E+00 5.0000E+01 0.0000E+00
246 1.0000E+00 3.2267E+03 1.1405E+00 5.3156E+00 2.8814E+01 1.0000E+00
247 6.6409E-02 3.4137E+03 1.1399E+00 5.8629E+00 4.9875E+01 9.0000E+00
248 7.4759E-02 3.4134E+03 1.1399E+00 5.8623E+00 4.9841E+01 8.0000E+00
249 8.5517E-02 3.4131E+03 1.1399E+00 5.8614E+00 4.9792E+01 7.0000E+00
250 9.9909E-02 3.4126E+03 1.1398E+00 5.8600E+00 4.9716E+01 6.0000E+00
251 1.2018E-01 3.4117E+03 1.1398E+00 5.8578E+00 4.9590E+01 5.0000E+00
252 1.3378E-01 3.4110E+03 1.1398E+00 5.8560E+00 4.9493E+01 4.5000E+00
253 1.5090E-01 3.4100E+03 1.1398E+00 5.8535E+00 4.9356E+01 4.0000E+00
254 1.7312E-01 3.4086E+03 1.1398E+00 5.8498E+00 4.9154E+01 3.5000E+00
255 2.0319E-01 3.4064E+03 1.1398E+00 5.8440E+00 4.8839E+01 3.0000E+00
256 2.4637E-01 3.4026E+03 1.1398E+00 5.8339E+00 4.8303E+01 2.5000E+00
257 3.1427E-01 3.3951E+03 1.1398E+00 5.8142E+00 4.7273E+01 2.0000E+00
258 4.4076E-01 3.3766E+03 1.1398E+00 5.7641E+00 4.4793E+01 1.5000E+00
259 9.9655E-01 3.2280E+03 1.1405E+00 5.3196E+00 2.8921E+01 1.0000E+00
260 1.0000E+00 3.2267E+03 1.1405E+00 5.3156E+00 2.8814E+01 1.0000E+00
261 2.6898E+00 2.2642E+03 1.1886E+00 2.5989E+00 1.7306E+00 5.0000E+00
262 3.1665E+00 1.9420E+03 1.2047E+00 2.3394E+00 6.7568E-01 1.0000E+01
263 3.6491E+00 1.6575E+03 1.2126E+00 2.2486E+00 2.6992E-01 2.0000E+01
264 3.9371E+00 1.5101E+03 1.2149E+00 2.2262E+00 1.5903E-01 3.0000E+01
265 4.1449E+00 1.4136E+03 1.2158E+00 2.2183E+00 1.0958E-01 4.0000E+01
266 4.3083E+00 1.3433E+03 1.2160E+00 2.2161E+00 8.2206E-02 5.0000E+01
267 4.4433E+00 1.2886E+03 1.2159E+00 2.2168E+00 6.5060E-02 6.0000E+01
268 4.5586E+00 1.2443E+03 1.2156E+00 2.2190E+00 5.3418E-02 7.0000E+01
269 4.6593E+00 1.2073E+03 1.2153E+00 2.2221E+00 4.5053E-02 8.0000E+01
270 4.7488E+00 1.1757E+03 1.2149E+00 2.2258E+00 3.8781E-02 9.0000E+01
```

```

271     4.8293E+00 1.1482E+03 1.2144E+00 2.2297E+00 3.3924E-02 1.0000E+02
272     4.9158E+00 1.1197E+03 1.2138E+00 2.2344E+00 2.9412E-02 1.1192E+02
273     4.9368E+00 1.1129E+03 1.2137E+00 2.2357E+00 2.8418E-02 1.1500E+02];
274
275 %Mach Number
276 M = [Data_Lower_Stage(:,1)';Data_Upper_Stage(:,1)'];
277
278 %Temperature in K
279 T = [Data_Lower_Stage(:,2)';Data_Upper_Stage(:,2)'];
280
281 %Gamma, ratio of specific heats
282 gam = [Data_Lower_Stage(:,3)';Data_Upper_Stage(:,3)'];
283
284 %Specific heat capacity at constant pressure, x103 (kilo, SI units)
285 Cp = [Data_Lower_Stage(:,4)';Data_Upper_Stage(:,4)'];
286
287 %Pressure of combustion gases, bar
288 P = [Data_Lower_Stage(:,5)';Data_Upper_Stage(:,5)'];
289
290 %Area ratio between considered section and throat
291 ae = [Data_Lower_Stage(:,6)';Data_Upper_Stage(:,6)'];
292
293
294 %% AXIAL VARIATIONS
295
296
297 %Nozzle Geometry
298
299 Ae_At= [15, 115]; %expansion ratio lower and upper stages
300 At = pi*Dt^2/4; %m^2, throat area
301 Ae = [At * Ae_At(1), At * Ae_At(2)]; %m^2, exit area
302 L_lower = 0.53; %m, length of nozzle
303 L_upper = 1.04; %m, length of nozzle
304
305
306 % r_g = Dt./2; %m, radius of throatr
307 % r_in = (Ae ./ pi).^0.5; %m, radius of nozzle inlet
308 % L_c = 0.1; %m, length of combustion chamber
309 % L_T = 0.5; %m, length of nozzle + combustion chamber
310 % r_l = @(x) r_g + (r_in(1)-r_g)*(1+cos(2*pi*(x-L_c)/L_n))/2; %lower stage
311 % r_u = @(x) r_g + (r_in(2)-r_g)*(1+cos(2*pi*(x-L_c)/L_n))/2; %upper stage
312 % % r goes from Lc to LT (combustion chamber not included in above eqn)
313
314 Dt = zeros;
315 r_val = zeros; %make function into array of values
316 %r_axis = linspace(L_c, L_T, length(M)); %create an array for x-axis
317 r_axis = linspace(0, 1, length(M)); %create an array for x-axis
318
319 location_throat = 0.5;
320 r_exit = [0.125 0.3461];
321 r_in = [0.0968 0.0968];
322 r_throat = 0.0323;
323
324 lsupersonic = @(x) r_throat + ((x-location_throat)).^0.5;
325 lsupersonic = @(x) (r_exit(1)/lsupersonic(1))....
326     *(r_throat + ((x-location_throat)).^0.5);
327

```



```

328 lsubsonic = @(x) r_throat + (location_throat-x).^0.5;
329 lsubsonic = @(x) (r_in(1)/lsubsonic(0))...
330     *(r_throat + (location_throat-x).^0.5);
331
332 usupersonic = @(x) r_throat + ((x-location_throat)).^0.5;
333 usupersonic = @(x) (r_exit(2)/usupersonic(1))...
334     *(r_throat + ((x-location_throat)).^0.5);
335
336 usubsonic = @(x) r_throat + (location_throat-x).^0.5;
337 usubsonic = @(x) (r_in(2)/usubsonic(0))...
338     *(r_throat + (location_throat-x).^0.5);
339
340 for i = 1:length(M)
341
342     if i <= 13
343         Dt(1,i) = 2*lsubsonic(i*location_throat/(0.5*length(M)));
344         Dt(2,i) = 2*usubsonic(i*location_throat/(0.5*length(M)));
345     end
346
347     if i >= 14
348         Dt(1,i) = 2*lsubsonic(location_throat + (i-13)...
349             *(1-location_throat)/(0.5*length(M)));
350         Dt(2,i) = 2*usupersonic(location_throat + (i-13)...
351             *(1-location_throat)/(0.5*length(M)));
352     end
353
354     % Dt(1,i) = 2 * r_l(L_c + i*(L_T-L_c)/length(M))';
355     % Dt(2,i) = 2 * r_u(L_c + i*(L_T-L_c)/length(M))';
356     % r_val = r_l(L_c + i*(L_T-L_c)/length(M))';
357
358 end
359
360 % 2 rows - first is lower and second is upper stage
361
362 %Set Gas Side Wall Temperature
363 %T_wg = linspace(1400,1000,length(M)); %assume
364 T_wg = linspace(600,500,length(M));
365
366 %Prandtl number of combustion gases
367 Pr_g = 4.*gam ./ (9.*gam -1 );
368
369 %Wall adiabatic temperature
370 T_aw = T.* (1 + Pr_g.^0.33.*M.^2.*(gam-1)/2) ./ (1 + M.^2.*(gam-1)./2);
371
372 %Correction factor for gas heat transfer parameter
373 sigma = (0.5.*(T_wg./T).*(1+M.^2.*(gam-1)/2)+0.5).^(-0.68...
374     .*(1+M.^2.*(gam-1)/2).^(-0.12);
375
376 %Bartz gas heat transfer parameter
377 h_g = (0.026./Dt.^0.2).*(Cp.*10^3*visc_g^0.2./Pr_g.^0.6)...
378     .*(P.*10^5*g/c).^0.8.*(ae).^(-0.9.*sigma);
379
380 %Heat transfer from hot gas side
381 q = abs(h_g .* (T_aw - T_wg));
382
383 %Total Heat Extracted
384 q_tot_lower_sub =...

```

```

385     (pi.*Dt(1,1:13).^2 /4).* q(1,1:13).*(0.09)/(0.5*length(M));
386 q_tot_upper_sub =...
387     (pi.*Dt(2,1:13).^2 /4).* q(2,1:13).*(0.09)/(0.5*length(M));
388
389 q_tot_lower_super =...
390     (pi.*Dt(1,14:end).^2 /4).* q(1,14:end).*(L_lower-0.09)/(0.5*length(M));
391 q_tot_upper_super =...
392     (pi.*Dt(2,14:end).^2 /4).* q(2,14:end).*(L_upper-0.09)/(0.5*length(M));
393
394 disp('Total Power Extracted From One Lower Stage Engine (KW): ');
395 disp((sum(q_tot_lower_sub(1,2:end))+sum(q_tot_lower_super(1,:)))*0.001);
396 disp('Total Power Extracted From One Upper Stage Engine (KW): ');
397 %subsonic region and 3 centimeters of super sonic region
398 disp((sum(q_tot_upper_sub(1,2:12))+sum(q_tot_upper_super(1,1)))*0.001);
399
400 %% AXIAL PLOTS
401
402 % figure
403 % sgtitle('LOWER STAGE VARIATIONS IN THE AXIAL DIRECTION',...
404 % 'Fontweight','bold','FontSize',20);
405 %
406 % tiledlayout(3,1)
407
408 figure
409
410 f = fit(r_axis(2:end)',h_g(1, 2:end)','smoothingspline');
411 plot(f,r_axis(2:end)',h_g(1, 2:end)');
412 grid on
413 ylabel('Heat Transfer Coefficient, W/(m^2K)','FontSize',20);
414 xlabel('Non-Dimensional Axial Distance','FontSize',20)
415 hold on
416 yyaxis right
417 ylabel('Nozzle Radial Distance, m','FontSize',20);
418 hold on
419 yyaxis right
420 fplot(lsupersonic, [0 1],'-')
421 ylim([0 1.5*r_exit(1)]);
422 hold on
423 fplot(lsubsonic, [0 1],'-');
424 legend('CEA Data','Fitted Curve','Nozzle');
425 title('Heat Transfer Coefficient','FontSize',20)
426
427 print('hgcoeff.png','-dpng','-r400');
428
429 figure
430
431 f = fit(r_axis',T_aw(1,:),'smoothingspline');
432 plot(f,r_axis',T_aw(1,:));
433 grid on
434 ylabel('Wall Adiabatic Temperature, K','FontSize',20);
435 xlabel('Non-Dimensional Axial Distance','FontSize',20)
436 %set(gca,'XTickLabel',[])
437 hold on
438 yyaxis right
439 ylabel('Nozzle Radial Distance, m','FontSize',20);
440 hold on
441 yyaxis right

```

```
442 fplot(lsupersonic, [0 1], '-')
443 ylim([0 1.5*r_exit(1)]);
444 hold on
445 fplot(lsubsonic, [0 1], '-');
446 legend('CEA Data', 'Fitted Curve', 'Nozzle');
447 title('Adiabatic Wall Temperature', 'FontSize', 20)
448
449 print('adiabbatictemp.png', '-dpng', '-r400');
450
451 figure
452
453 f=fit(r_axis(2:end)', q(1,2:end)', 'smoothing spline');
454 plot(f, r_axis(2:end)', q(1,2:end)');
455 grid on
456 ylabel('Heat Transfer, W/m^2', 'FontSize', 20);
457 xlabel('Non-Dimensional Axial Distance', 'FontSize', 20)
458 %set(gca, 'XTickLabel', [])
459 hold on
460 yyaxis right
461 ylabel('Nozzle Radial Distance, m', 'FontSize', 20);
462 hold on
463 yyaxis right
464 fplot(lsupersonic, [0 1], '-')
465 ylim([0 1.5*r_exit(1)]);
466 hold on
467 fplot(lsubsonic, [0 1], '-');
468 legend('CEA Data', 'Fitted Curve', 'Nozzle');
469 title('Heat Flux', 'FontSize', 20)
470
471 print('heattransfer.png', '-dpng', '-r400');
472
473
474 % figure
475 % sgtitle('UPPER STAGE VARIATIONS IN THE AXIAL DIRECTION', ...
476 % 'Fontweight', 'bold', 'FontSize', 20);
477 %
478 % tiledlayout(3,1)
479 %
480 % nexttile
481 %
482 % f = fit(r_axis(2:end)', h_g(2, 2:end)', 'smoothing spline');
483 % plot(f, r_axis(2:end)', h_g(2, 2:end)');
484 % grid on
485 % ylabel('Heat Transfer Coefficient, W/(m^2K)', 'FontSize', 15);
486 % xlabel('Non-Dimensional Axial Distance', 'FontSize', 15)
487 % hold on
488 % yyaxis right
489 % ylabel('Nozzle Radial Distance, m', 'FontSize', 15);
490 % hold on
491 % yyaxis right
492 % fplot(usupersonic, [0 1], '-')
493 % ylim([0 1.5*r_exit(2)]);
494 % hold on
495 % fplot(usubsonic, [0 1], '-');
496 % legend('CEA Data', 'Fitted Curve', 'Nozzle');
497 %
498 % nexttile
```

```
499 %
500 % f = fit(r_axis',T_aw(2,:)','smoothingspline');
501 % plot(f,r_axis',T_aw(2,:)');
502 % grid on
503 % ylabel('Wall Adiabatic Temperature, K','FontSize',15);
504 % xlabel('Non-Dimensional Axial Distance','FontSize',15)
505 % hold on
506 % yyaxis right
507 % ylabel('Nozzle Radial Distance, m','FontSize',15);
508 % hold on
509 % yyaxis right
510 % fplot(usupersonic, [0 1],'-')
511 % ylim([0 1.5*r_exit(2)]);
512 % hold on
513 % fplot(usubsonic, [0 1],'-');
514 % legend('CEA Data','Fitted Curve','Nozzle');
515 %
516 % nexttile
517 %
518 % f=fit(r_axis(2:end)',q(2,2:end)','smoothingspline');
519 % plot(f,r_axis(2:end)',q(2,2:end)');
520 % grid on
521 % ylabel('Heat Transfer, W/m^2','FontSize',15);
522 % xlabel('Non-Dimensional Axial Distance','FontSize',15)
523 % hold on
524 % yyaxis right
525 % ylabel('Nozzle Radial Distance, m','FontSize',15);
526 % hold on
527 % yyaxis right
528 % fplot(usupersonic, [0 1],'-')
529 % ylim([0 1.5*r_exit(2)]);
530 % hold on
531 % fplot(usubsonic, [0 1],'-');
532 % legend('CEA Data','Fitted Curve','Nozzle');
533
534 %% FRICTION
535
536 %Density variations for supercritical Methane
537
538 density_10_bar = ...
539     [100 100 446.0
540      200 100 266.2
541      240 100 128.4
542      260 100 101.0
543      280 100 85.51
544      300 100 75.17
545      320 100 67.61
546      340 100 61.75
547      360 100 57.01
548      400 100 49.73
549      500 100 38.32
550      600 100 31.47];
551
552 density_5_bar = ...
553     [100 50 442.5
554      200 50 87.76
555      250 50 50.00
```

```

556     300 50 34.97
557     350 50 28.00
558     400 50 24.60
559     500 50 19.27
560     600 50 15.93];
561
562 %Plot density variations for supercritical methane
563
564 figure
565 f1=fit(density_10_bar(:,1),density_10_bar(:,3),'smoothingspline');
566 plot(f1,'r');
567 title('Density Variations for Supercritical Methane','FontSize',20);
568 grid on
569 hold on
570 plot(density_10_bar(:,1),density_10_bar(:,3),'rx');
571 hold on
572 f2=fit(density_5_bar(:,1),density_5_bar(:,3),'smoothingspline');
573 plot(f2,'b');
574 hold on
575 plot(density_5_bar(:,1),density_5_bar(:,3),'bx');
576 hold off
577 xlabel('Temperature, K','FontSize',20);
578 ylabel('Density, kg/m^3','FontSize',20);
579 legend('10 bar Data', '10 bar fit curve','5 bar Data', '5 bar fit curve');
580
581 print('density.png','-dpng','-r400');
582
583 T_co = linspace(200,100,length(M)); %set coolant temperature (user defined)
584 rho = f2(T_co);
585
586 N = 31; %pick number of channels
587 d = (pi*Dt + 1.6*t(2)*pi - 2*t(2)*N) ./ (N - 0.8*pi);
588
589 Re = 4*m_f(2) ./ (visc_c * pi * N .* d); %mass flow of 10kg/s
590
591 h = 0.000001; %channel roughness height
592
593 fric_func = zeros;
594 Dp = zeros;
595 fric_coeff_val = zeros;
596 L = (L_lower)/length(M); %length of one section considered
597 u = zeros;
598
599 for i = 1:length(M)
600
601     for j = 1:2 %stages
602
603         %find friction coefficient iteratively
604         for fric = 0.001:0.001:10
605
606             %Colebrook Equation
607             fric_func = 1/fric^0.5...
608                 + 2*log10(h/(3.7.*d(j,i)) + 2.51/(Re(j,i).*fric^0.5));
609
610             if fric_func > -0.5 && fric_func < 0.5
611                 break
612             end

```

```

613
614     end
615
616     %disp(fric);
617
618     fric_coeff_val(j,i) = fric;
619
620     u(j,i) = 4 * m_f(2) / (rho(i)*pi*N*d(j,i)^2);
621
622     Dp(j,i) = 0.5*fric_coeff_val(j,i)*L*rho(i)* u(j,i)^2 / d(j,i);
623
624     end
625
626 end
627
628 figure
629
630 f=fit(r_axis(1:end)',Dp(1,1:end)','smoothingspline');
631 plot(f,r_axis(1:end)',Dp(1,1:end)');
632 grid on
633 ylabel('Pressure Drop, Pa','FontSize',20);
634 xlabel('Non-Dimensional Axial Distance','FontSize',20);
635 title('Pressure Drop at Every Axial Section','FontSize',20);
636 hold on
637 yyaxis right
638 ylabel('Nozzle Radial Distance, m','FontSize',20);
639 hold on
640 yyaxis right
641 fplot(lsupersonic, [0 1],'-')
642 ylim([0 0.2]);
643 hold on
644 fplot(lsubsonic, [0 1],'-');
645 legend('Pressure Drop','Fitted Curve','Nozzle');
646 fun_pressure = @(x) feval(f,x); %pressure drop at every axial section
647 %total_area = integral(fun_pressure,L_c,L_T,'ArrayValued',true);
648
649 print('pressuredrop.png','-dpng','-r400');
650
651 figure
652
653 f = fit(r_axis(1:end)',cumsum(Dp(1,1:end),'reverse'),'smoothingspline');
654 plot(f,r_axis(1:end)',cumsum(Dp(1,1:end),'reverse'));
655 grid on
656 ylabel('Cumulative Pressure Drop, Pa','FontSize',20);
657 xlabel('Non-Dimensional Axial Distance','FontSize',20);
658 title('Cumulative Pressure Drop','FontSize',20);
659 hold on
660 yyaxis right
661 ylabel('Nozzle Radial Distance, m','FontSize',20);
662 hold on
663 yyaxis right
664 fplot(lsupersonic, [0 1],'-')
665 ylim([0 0.2]);
666 hold on
667 fplot(lsubsonic, [0 1],'-');
668 legend('Cumulative Pressure Drop','Fitted Curve','Nozzle');
669

```

```
670 print('cumulativepressure.png','-dpng','-r400');
671
672 total_pressure_drop = sum(Dp(1,:));
673 disp('Total Pressure Drop Across Cooling Jacket (bar): ');
674 disp(total_pressure_drop/10^5);
675
676
677 %% PRESSURIZATION
678
679 %Calculation to determine total mass of Helium
680
681 %FIRST VALUE IS FOR OXIDIZER AND SECOND FOR METHANE TANK
682
683 %Propellant storage temps halfway between melting and boiling points
684 T_prop = [(90.19+54.36)/2 (111.65+90.7)/2];
685
686 %experimental heat transfer coeff between liquid and gas prop
687 %based on 0.002BBtu/s ft^2 R from Modern Engineering Design book
688 H = [1.6299 1.6299];
689
690 O_F = 2.95;
691 T_ullage = [120 120]; %assumed, K
692 Cp_l = [0.347 3.358]*10^3; %specific heat of liquid prop
693 Cp_v = [0.915 2.232]*10^3; %specific heat of vapour prop (J/kgK)
694 T_v = [90.19 111.65]; %vaporization temp of prop
695 h_v = [214 511]*10^3; %latent heat of vaporization of prop (J/kg)
696 rho_tanks = [1141 422]; %kg/m^3, density of propellants
697 R_prop = [259.8 518.3]; %specific gas constant of propellants J/kgK
698
699 %FIRST VAL IS FOR LOWER STAGE AND SECOND FOR UPPER STAGE
700 m_flow_rate = [8.2*9; 8.2]; %kg/s
701 m_flow_rate_ox = (O_F / (O_F + 1))*m_flow_rate;
702 m_flow_rate_meth = (1 / (O_F + 1))*m_flow_rate;
703 m_flow_rate = [m_flow_rate_ox m_flow_rate_meth];
704
705 m_prop = [8250; 1578.2]; %kg
706 m_ox = (O_F / (O_F + 1))* m_prop;
707 m_meth = (1 / (O_F + 1))* m_prop;
708
709 m_prop = [m_ox m_meth]; %total propellant mass
710 t_burn_out = m_prop ./ m_flow_rate; %s, burn out time
711 V_tank = m_prop./rho_tanks;
712 A_tank = 0.79; %area of tanks, m^2
713 P_tank = 3*10^5; %pressure of tanks
714 R_pressurant = 2077; %specific gas constant value for pressurant
715
716 % Q = HAt(Tu - Tp)
717 % Q = Mv[(Cp_l(Tv-Tp) + h_v + Cp_v(Tu-Tv))]
718
719 num = H * A_tank .* t_burn_out .* (T_ullage - T_prop);
720 den = Cp_l.*(T_v-T_prop) + h_v + Cp_v.*(T_ullage-T_v);
721
722 M_v = num ./ den;
723
724 V_v = M_v .* R_prop .* T_ullage ./ P_tank;
725 V_pressurant = V_tank - V_v;
726 M_pressurant = P_tank .* V_pressurant ./ (R_pressurant.*T_ullage);
```

```

727 disp('Helium mass lower stage, for oxidizer tank (kg)');
728 disp(M_pressurant(1,1));
729 disp('Helium mass lower stage, for methane tank (kg)');
730 disp(M_pressurant(1,2));
731 disp('Helium mass upper stage, for oxidizer tank (kg)');
732 disp(M_pressurant(2,1));
733 disp('Helium mass upper stage, for methane tank (kg)');
734 disp(M_pressurant(2,2));
735 disp('Total Helium Required, (kg)');
736 disp(sum(M_pressurant(:,1)) + sum(M_pressurant(:,2)));
737
738
739 %% VALVES
740
741 T_valve = 300; %temperature of coolant at position where valve is
742 Q = m_flow_rate_meth(2)*3600 / f2(T_valve); %flow rate, m^3/h (f2 is rho)
743 SG = 0.5537; %specific gravity of fluid
744
745 syms Delta_P %pressure drop across valve in bar
746 K_v = @(Delta_P) Delta_P.^-0.5 * Q * SG^0.5; %flow factor in m^3/h/bar^0.5
747
748 K_v_values = [46,72,105,170,275]; %Kv values for ball valve
749 orifice = [20,25,32,40,50]; %orifice diameter for ball valve
750 Pressure = SG./ ((K_v_values./Q).^2);
751
752 figure
753
754 hold on
755 for i = 1:length(K_v_values)
756     plot([0 Pressure(i)], [K_v_values(i) K_v_values(i)], 'b', 'Linewidth', 0.5);
757     plot([Pressure(i) Pressure(i)], [0 K_v_values(i)], 'b', 'Linewidth', 0.5);
758     pressure_label = [Pressure(i) K_v_values(i)];
759     text(Pressure(i), K_v_values(i), num2str(orifice(i)), 'fontsize', 15);
760 end
761 hold on
762
763 text(9, 550, 'Orifice diameters, mm', 'Edgecolor', ...
764     'k', 'backgroundcolor', 'w', 'fontsize', 15);
765
766
767 fplot(K_v, 'r', 'Linewidth', 1)
768 xlabel('Pressure Drop Across Ball Valve, bar', 'FontSize', 20)
769 ylabel('K_v Flow Coefficient m^3/(hr bar^{0.5})', 'FontSize', 20)
770 %set(gca, 'YScale', 'log')
771 title('Flow Coefficient vs Pressure Drop for a Ball Valve', 'FontSize', 20)
772 grid on
773
774 print('ballvalve.png', '-dpng', '-r400');
775
776 K_v_values_globe = [12, 17.8, 23.8, 37, 65];
777 orifice_globe = [25, 32, 40, 50, 65];
778 Pressure_globe = SG./ ((K_v_values_globe./Q).^2);
779
780 figure
781
782 hold on
783 for i = 1:length(K_v_values_globe)

```



```

784     plot([0 Pressure_globe(i)], [K_v_values_globe(i) K_v_values_globe(i)], 'b', 'Linewidth', 0.5);
785     plot([Pressure_globe(i) Pressure_globe(i)], [0 K_v_values_globe(i)], 'b', 'Linewidth', 0.5);
786     pressure_label = [Pressure_globe(i) K_v_values_globe(i)];
787     text(Pressure_globe(i), K_v_values_globe(i), num2str(orifice_globe(i)), 'fontsize', 15);
788 end
789 hold on
790
791 text(150, 85, 'Orifice diameters, mm', 'Edgecolor', ...
792      'k', 'backgroundcolor', 'w', 'fontsize', 15);
793
794
795 fplot(K_v, 'r', 'Linewidth', 1)
796 xlabel('Pressure Drop Across Globe Valve, bar', 'FontSize', 20)
797 ylabel('K_v Flow Coefficient m^3/(hr bar^{0.5})', 'FontSize', 20)
798 ylim([0 100])
799 %set(gca, 'YScale', 'log')
800 title('Flow Coefficient vs Pressure Drop for a Globe Valve', 'FontSize', 20)
801 grid on
802
803 print('globevalve.png', '-dpng', '-r400');
804
805 %% LINES
806
807 % Data for diameter of lines from NASA in mm
808 fscale1 = 1;
809 pressurization = fscale1*[22.225, 31.75, 12.7, 15.875, 12.7, 23.8125,...
810      22.225, 12.7, 31.75, 9.525, 9.525, 19.06, 16.383, 15.875, 19.05];
811 bypass = fscale1*[22.225, 31.75, 12.7, 15.875, 12.7, 12.7, 31.75,...
812      9.525, 9.525, 16.383, 15.875];
813 hydraulic = fscale1*[22.225, 31.75, 12.7, 15.875, 12.7, 12.7, 31.75,...
814      9.525, 9.525, 16.383, 15.875];
815
816 fscale2 = 0.5;
817 pump_discharge = fscale2*[132.08, 86.36, 93.98, 120.65, 101.6];
818 pump_inlet = fscale2*[50.8, 149.352, 174.752, 58.42];
819
820 bscale = 0.2;
821 bellow_hydraulic_sys = bscale * [231.775, 169.799, 101.6, 101.6, 203.2, 203.2];
822 bellow_pressurization = bscale * [31.75, 34.93, 38.1, 76.2, 76.2, 38.1, 76.2, 76.2];
823 bellow_turbine = bscale * [685.8, 457.2, 373.38, 203.3, 203.3, 203.3];
824
825 % Mean diameters
826 mean_pressurization = num2str(round(mean(pressurization), 2));
827 mean_bypass = num2str(round(mean(bypass), 2));
828 mean_hydraulic = num2str(round(mean(hydraulic), 2));
829 mean_pump_discharge = num2str(round(mean(pump_discharge), 2));
830 mean_pump_inlet = num2str(round(mean(pump_inlet), 2));
831 mean_bellow_hydraulic_sys = num2str(round(mean(bellow_hydraulic_sys), 2));
832 mean_bellow_pressurization = num2str(round(mean(bellow_pressurization), 2));
833 mean_bellow_turbine = num2str(round(mean(bellow_turbine), 2));
834
835 data_small_lines = ...
836     {pressurization, bypass, hydraulic, pump_discharge, pump_inlet};
837 dataLabels_small_lines = ...
838     {'Pressurisation', 'Bypass', 'Hydraulic', 'Pump Outlet', 'Pump Inlet'};
839
840 data_bellow = ...

```

```
841     {bellow_hydraulic_sys,bellow_pressurization,bellow_turbine};
842 dataLabels_bellow =...
843     {'Hydraulic', 'Pressurisation', 'Turbine Exhaust and Pump Inlet'};
844
845 % tiledlayout(1,2)
846 % nexttile
847
848 % plotting scatter plot
849 figure
850 hold on;
851 for i = 1:numel(data_small_lines)
852     if mod(i,2) == 0
853         scatter(repmat(i,1,length(data_small_lines{i})),...
854             data_small_lines{i},'+','LineWidth',1.5);
855     else
856         scatter(repmat(i,1,length(data_small_lines{i})),...
857             data_small_lines{i},'o','LineWidth',1.5);
858     end
859 end
860 grid on
861 hold off;
862
863 title('Flex Hose Diameters','FontSize',20);
864 ylabel('Diameter, mm','FontSize',20);
865 xlabel('Engine Section','FontSize',20)
866
867 ax = gca;
868 ax.XTick = 1:numel(dataLabels_small_lines);
869 ax.XTickLabel = dataLabels_small_lines;
870 ax.XLim = [0 numel(dataLabels_small_lines)+1];
871 lgd = legend(mean_pressurization,mean_bypass,mean_hydraulic,...
872     mean_pump_discharge,mean_pump_inlet,'Location','northwest');
873 title(lgd,'Mean Values','FontSize',12)
874
875 print('flexhose.png','-dpng','-r400');
876
877 %nexttile
878 % plotting scatter plot
879 figure
880 hold on;
881 for i = 1:numel(data_bellow)
882     if mod(i,2) ==0
883         scatter(repmat(i,1,length(data_bellow{i})),...
884             data_bellow{i},'+','LineWidth',1.5);
885     else
886         scatter(repmat(i,1,length(data_bellow{i})),...
887             data_bellow{i},'o','LineWidth',1.5);
888     end
889
890 end
891 grid on
892 hold off;
893
894 title('Bellow Diameters','FontSize',20);
895 ylabel('Diameter, mm','FontSize',20);
896 xlabel('Engine Section','FontSize',20)
897
```

```

898 ax = gca;
899 ax.XTick = 1:numel(dataLabels_bellow);
900 ax.XTickLabel = dataLabels_bellow;
901 ax.XLim = [0 numel(dataLabels_bellow)+1];
902 lgd = legend(mean_bellow_hydraulic_sys,...
903     mean_bellow_pressurization,mean_bellow_turbine,'Location','northwest');
904 title(lgd,'Mean Values','FontSize',12)
905
906 print('bellow.png','-dpng','-r400');
907
908 %% NUMBER OF STAGES FOR ROCKET
909
910 %plots graph of G vs ?V (G = lift off mass to payload mass)
911
912 ge = 9.81; %m/s^2
913 Isp = 300; %s
914 epsilon = 0.1; %stage structural factor mf/m0
915
916 syms G
917
918 N_stages = (1:5); %number of stages
919 f = - N_stages * ge * Isp .* log(epsilon*(1 - G.^(-1./N_stages))...
920     + G.^(-1./N_stages)); %rocket equation
921
922 figure
923
924 fplot(f,G,[0 2000])
925 title('Number of Stages Required', 'FontSize', 20);
926 ylabel('G ratio', 'FontSize', 20);
927 xlabel('Delta V, m/s', 'FontSize', 20);
928 grid on
929
930 Payload_mass = 160; %kg
931 Lift_off_mass = 18924; %kg
932 G_rocket = Lift_off_mass / Payload_mass;
933 disp('G = ');
934 disp(G_rocket);
935 Delta_v = 10006;
936
937 hold on
938 plot([0 Delta_v],[G_rocket G_rocket],'color','r','linewidth',1);
939 plot([Delta_v Delta_v],[0 G_rocket],'color','r','linewidth',1);
940 xlim([0 14000])
941 ylim([0 300])
942 legend('N = 1','N = 2','N = 3', 'N = 4', 'N = 5',...
943     'Design G','location','northwest')
944 hold off
945
946 print('numberofstages.png','-dpng','-r400');
947
948 %% METHANE SUPERCRITICAL DIAGRAM
949
950
951 critical_temp = 190.56; %K
952 critical_pressure = 45.992; %MPa
953 melting_temp = 90.7; %K
954

```

```
955 scatter(critical_temp,critical_pressure)
956 grid on
957 text(critical_temp + 5,critical_pressure + 5, ...
958     'Critical Point','FontSize',20)
959 text(critical_temp + 20,critical_pressure + 20,...
960     'Supercritical Phase','FontSize',12)
961 text(critical_temp + 20,critical_pressure - 20,...
962     'Gaseous Phase','FontSize',12)
963 text(critical_temp - 70,critical_pressure + 20,...
964     'Compressible Liquid','FontSize',12)
965 text(critical_temp - 70,critical_pressure - 20,...
966     'Liquid Phase','FontSize',12)
967 text(melting_temp - 35,critical_pressure,'Solid Phase','FontSize',12)
968 hold on
969 plot([melting_temp melting_temp],[0 100],'Linewidth',1)
970 hold on
971 plot([melting_temp 273.15],...
972     [critical_pressure critical_pressure],'Linewidth',1)
973 hold on
974 plot([critical_temp critical_temp],[critical_pressure 100],'Linewidth',1)
975 hold on
976 syms x
977 y = 0.004612*x^2 - 0.8366*x + 37.94;
978 fplot(y,[melting_temp critical_temp])
979 ylim([0 100])
980 xlim([50 273.15])
981 xlabel('Temperature, K','FontSize',20)
982 ylabel('Pressure, bar','FontSize',20)
983 title('Methane Phase Diagram','FontSize',20)
984
985 print('methanephase.png','-dpng','-r400');
```

Bibliography

- [1] Granta Design Limited. *CES EduPack software*, 2009.
- [2] Dieter K. Huzel and David H. Huang. *Liquid Rocket Lines, Bellows Flexible Hoses, and Filters*, volume NASA SP-8123. NASA Astrophysics Data System, Washington, D.C., 1977.
- [3] United Performance Metals. *Stainless 321. Quality Solution. Trusted Partners*, 2020.
- [4] United Performance Metals. *Inconel 718. Quality Solution. Trusted Partners*, 2020.
- [5] Engineering toolbox, 2008. URL: https://www.engineeringtoolbox.com/methane-d_1420.html/.
- [6] R. Joseph Cassady Stephen D. Heister, William E. Anderson, Timothée L. Pourpoint. *Rocket Propulsion*. Cambridge Aerospace Series. Cambridge University Press, 2019.
- [7] Elon Musk. SpaceX™, 2002. URL: <https://www.spacex.com>.
- [8] Peter Beck. RocketLabUSA™, 2013. URL: <https://www.rocketlabusa.com>.
- [9] Joe Pappalardo. The rocket fuel rivalry shaping the future of spaceflight, 2018. URL: <https://www.popularmechanics.com/space/rockets/>.
- [10] Dieter K. Huzel and David H. Huiang. *Design of Liquid Propellant Rocket Engines*. NASA, Scientific and Technical Information Division, Washington D. C., second edition, 1967.
- [11] Dr.-Ing. Peter H. Weutas, Dipl.-Ing. Markus Ortelt, Dipl.-Ing. Thoralf Hofmann. *Development of reusable liquid propulsion units based on transpiration cooled, ceramic thrust chambers and turbo pumps for launch and in space applications*. Glasgow, UK, 2017.
- [12] George P. Sutton, Oscar Biblarz. *Rocket Propulsion Elements*. Proceedings of the Royal Society A. John Wiley and Sons, 8th revised edition edition, 2010.
- [13] Tokio Nara, Mayuki Niitsu, Akihiro Sato, Tadaoki Onga, Junya Takida, Nobuki Negoro. *Development Status of H3 Launch Vehicle*, volume 2. JAXA, Mitsubishi Heavy Industries, 2017.
- [14] Holger Burkhardt, Martin Sippel, Armin Herbertz, Josef Klevanski. *Comparative Study of Kerosene and Methane Propellant Engines for Reusable Liquid Booster Stages*. DLR, German Aerospace Center, Deutsches Zentrum für Luft- und Raumfahrt, 51147 Köln, Germany, 2002.
- [15] Dieter K. Huzel and David H. Huang. *Modern Engineering For Engineering Design of Liquid-Propellant Rocket Engines*, volume 147. American Institute of Aeronautics and Astronautics, 370 L’Enfant Promenade, SW, Washington D.C., 20024-1528, 1992.
- [16] C. F. Colebrook and C. M. White. *Experiments with fluid friction in roughened pipes*, volume 161 of *Proceedings of the Royal Society A*. Royal Society, 1937.
- [17] Glenn O. Brown. *The History of the Darcy-Weisbach Equation for Pipe Flow Resistance*. Academic Press, Stillwater, OK 74078, 2002. doi:10.1061/40650(2003)4.
- [18] S. Gordon and B. J. McBride. *Computer program for calculation of complex chemical equilibrium compositions and applications*. NASA, 1994.
- [19] Marco Pizzarelli. *Regenerative cooling of liquid rocket engine thrust chambers*. Agenzia Spaziale Italiana, 2017.

- [20] D.R. Bartz. *Turbulent boundary-layer heat transfer from rapidly accelerating flow of rocket combustion gases and of heated air*, volume 2 of *Advances in Heat Transfer*. Academic Press Inc. Elsevier Ltd, 370 L'Enfant Promenade, SW, Washington D.C., 20024-1528, 1965.
- [21] ACMA Staff. Pros and cons of additive manufacturing, 2014. URL: <http://compositesmanufacturingmagazine.com/2014/10/pros-cons-additive-manufacturing/>.
- [22] Hiroya Asakawa, Masahiro Tanaka, Takeshi Takaki and Kazuhiro Higashi. *Component tests of a LOX/methane full-expander cycle rocket engine: Electrically actuated valve*. JAXA, Hiroshima University, 2019. doi:10.13009/EUCASS2019-222.
- [23] A. Bhatia. *Control Valve Basics: Sizing and Selection*, volume M04-034. Continuing Education and Development, Inc., 9 Greyridge Farm Court Stony Point, NY 10980, 2014.
- [24] Christian Bürkert GmbH and Co. KG. *2/2 and 3/2-way all valve with pneumatic rotary actuator*. Fine Controls (UK) LTD, Wirral, CH62 3QL UK, 2020.
- [25] Christian Bürkert GmbH and Co. KG. *Pneumatic operated 2-way Globe Control Valve ELEMENT*. Fine Controls (UK) LTD, Wirral, CH62 3QL UK, 2020.
- [26] SpiraxSarco. Knuth: Computers and typesetting, pipeline ancilleries, check valves, 2020. URL: <https://www.spiraxsarco.com/learn-about-steam/pipeline-ancillaries/check-valves>.
- [27] J.C. Lee and P. Ramirez. *Pressurization Systems for Liquid Rockets*. NASA, Scientific and Technical Information Division, Washington D. C., 1975.